The pair singular series of a polynomial: an exact identity, its Dedekind zeta function, and the variance of prime values

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Abstract

How many primes lie among the values $f(1), f(2), f(3), \ldots$ of a polynomial, and how much does that number fluctuate? The first question has a conjectural answer, the Bateman–Horn constant $C(f)$; this paper is about the second, which for the sequence of all primes is the question that leads to the zeros of the Riemann zeta function. The fluctuation is governed by an explicit arithmetic function, the pair singular series $S_f(h)$, the Euler product that under the Hardy–Littlewood conjecture weights the simultaneous primality of $f(t)$ and $f(t+h)$; all our theorems concern S_f and involve no prime. We prove an exact identity for $\sum_{h\le H}(S_f(h) - C(f)^2)$ in terms of the roots of f modulo squarefree d; that its diagonal part has Dirichlet series $\zeta_K(s+1)E_f(s)$, $K = \mathbb{Q}[t]/(f)$, with residue exactly $1/C(f)$ and $1/E_f$ essentially the Artin L-function of the symmetric square of the permutation representation on the roots, so that the Dedekind zeta function of the field cut out by f governs the second moment of its prime values; a closed formula for $S_f(h)$ when f is quadratic; and that the Cesàro form of

$$\sum_{h\le H}\big(S_f(h) - C(f)^2\big) = -\tfrac{1}{2}\, C(f)\log H + O_f(1)$$

holds unconditionally for linear f and, for general f, exactly when the sum over pairs of distinct roots is $o(H\log H)$. We conjecture the asymptotic for every f, with the *first* power of $C(f)$ rather than the second that a naive rescaling predicts; 25 quadratics computed exactly to $H = 10^6$ give the exponent 0.87 ± 0.09. In plain terms: primes along any polynomial sequence fluctuate less than independent coin flips would, by the mechanism that Montgomery and Soundararajan found for primes in short intervals, and the size of the effect is set by an L-function attached to the polynomial. Assuming the Hardy–Littlewood pair conjecture, the law $\operatorname{Var}/\mathbb{E} \to 1 - \log T/\log X$ therefore holds for the prime values of every polynomial, with a polynomial-dependent correction that is large at accessible T, as prime counts along 67 polynomial sequences confirm against a random-integer control.

1 Introduction

For an irreducible polynomial $f \in \mathbb{Q}[t]$ that is integer-valued, has positive leading coefficient and no fixed prime divisor, the Bateman–Horn conjecture [4] (Conjecture F of Hardy and Littlewood [12] for quadratics; see [1] for a survey) predicts

$$\#\{t < T :\ f(t) \text{ prime}\} \sim C(f)\sum_{t<T}\frac{1}{\log f(t)}, \qquad C(f) = \prod_p \frac{1-\omega_f(p)/p}{1-1/p}, \tag{1}$$

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†Anthropic large language model (`claude-fable-5-1`). See the statement on authorship and AI use at the end. Most journals do not permit an AI system as an author; if the venue requires it, this line becomes an acknowledgement and the statement the required disclosure.

where $\omega_f(p)$ is the number of roots of f modulo p (for primes dividing the denominator of f, $\omega_f(p)/p$ denotes the density of t with $p \mid f(t)$). The object of this paper is not the count of primes but an arithmetic function attached to f. For $h \geq 1$ let

$$S_f(h) = \prod_p \frac{1 - \omega_{f,2}(p,h)/p}{(1-1/p)^2}, \qquad \omega_{f,2}(p,h) = 2\omega_f(p) - \nu_f(p,h), \tag{2}$$

where $\nu_f(p,h) = \#\{t \bmod p : \ p \mid f(t) \text{ and } p \mid f(t+h)\}$ is the number of roots s of f modulo p such that $s+h$ is again a root (with the same convention for primes dividing the denominator). This is the *pair singular series* of Hardy and Littlewood: heuristically, $S_f(h)/(\log f(t) \log f(t+h))$ is the probability that $f(t)$ and $f(t+h)$ are both prime, and for $f(t) = t$ it is the twin-prime singular series $\mathfrak{S}(h)$. But $S_f(h)$ is also, in itself, an explicit and exactly computable function of h: an absolutely convergent Euler product, equal to $C(f)^2$ times a product over the primes dividing a resultant, and for quadratics a finite product of twin-prime type (Theorem 5). Everything we prove concerns this function and no prime. Its mean value is $C(f)^2$ (Theorem 2 gives $\sum_{h \leq H} S_f(h) = C(f)^2 H + O(\log H)$), and the question is the second-order term. Our main proposal is:

Conjecture 1. *For every irreducible, integer-valued f without fixed prime divisor,*

$$\sum_{h \leq H} \bigl(S_f(h) - C(f)^2\bigr) = -\tfrac{1}{2}\, C(f) \log H + A_f + o(1) \qquad (H \to \infty), \tag{3}$$

with a constant A_f depending on f.

For $f(t) = t$ this is a theorem: $\sum_{h \leq H} \mathfrak{S}(h) = H - \frac{1}{2}\log H + O((\log H)^{2/3})$, due to Friedlander and Goldston [7] and Vaughan [23] after Montgomery and Soundararajan [18], who derived from it, assuming a uniform Hardy–Littlewood conjecture, that the number of primes in an interval of length H near X has

$$\frac{\mathrm{Var}}{\mathbb{E}} \approx 1 - \frac{\log H}{\log X}. \tag{4}$$

(Gallagher [8] had shown that the prime-tuple conjectures imply Poisson statistics in intervals of length $\lambda \log X$, and Goldston and Montgomery [9] that, under the Riemann Hypothesis, the asymptotic variance for $H = X^\theta$ is equivalent to Montgomery's pair correlation conjecture for the zeros of ζ [17]; in the function-field setting the variance statements are theorems [15].) The point of Conjecture 1 is the *first* power of $C(f)$: the second-order term is not $C(f)^2$ times the linear case, as a naive rescaling of the twin-prime heuristic would give, but $C(f)$ times it.

Why this matters, in plain terms. A polynomial sequence such as $t^2 + 1$ is expected to contain primes with density $C(f)/\log f(t)$, as if each value were an independent coin flip with that probability. Independent coin flips would make the number of primes among the first T values fluctuate with variance equal to its mean. Primes are not independent: whether $f(t)$ is divisible by a small prime p repeats with period p in t, so the sieving by small primes is deterministic along the sequence and only the large primes contribute randomness. The function $S_f(h)$ measures the resulting correlation between $f(t)$ and $f(t+h)$, and the sum $\sum_{h \leq H}(S_f(h) - C^2)$ is the total correlation up to distance H: it is negative, and it is exactly the amount by which the variance falls short of the coin-flip value. Our result is that this shortfall is $\frac{1}{2}C(f)\log H$ for every polynomial, with the *first* power of $C(f)$, and that it is generated by the Dedekind zeta function of the number field defined by f. For the linear polynomial this is the classical statement, going back to Montgomery and Soundararajan, that primes in short intervals are "more regular than random"; the present paper shows that the same regularity, with the same constant $\frac{1}{2}$, holds along every polynomial, and identifies the L-function behind it. What the paper does *not* do is prove anything about the primes themselves: the translation from S_f to prime counts assumes the Hardy–Littlewood pair conjecture, which is open for every non-linear f.

Results. Section 3 proves an exact identity for the left side of (3) (Theorem 2), identifies the Dirichlet series of its diagonal part as $\zeta_K(s+1)E_f(s)$ with residue $1/C(f)$ at $s=0$ (Theorem 3), which is where the first power of $C(f)$ comes from, identifies E_f as the inverse of the Artin L-function of the symmetric square of the permutation representation on the roots (Theorem 4), gives a closed twin-prime-type formula for $S_f(h)$ when f is quadratic (Theorem 5), proves the Cesàro form of (3) for every linear f and reduces it for general f to a bound $o(H\log H)$ for the off-diagonal sum over pairs of distinct roots of f modulo d (Theorem 6; the constant term needs the sharper Hypothesis (E)), tests the conjecture numerically for 110 polynomials of degree 1–4 up to $H=10^5$ and, exactly and with $C(f)$ varying over more than a decade, for 25 quadratics up to $H=10^6$ (Tables 1, 2, Figures 1, 2), isolates the off-diagonal remainder for quadratics (Table 3), and proves that over $\mathbb{F}_q[u]$ the off-diagonal contribution of every modulus of degree $\leq N$ vanishes identically, the remainder being a finite sum over moduli of degree between N and about $3N$ (Theorem 8, Lemma 9). Section 4 draws a consequence: for a quadratic the diagonal Riesz means have an explicit formula over the zeros of $\zeta^2L(\chi_D)$ (Proposition 10), explains why these zeros are nevertheless invisible in a direct computation to $4\cdot 10^7$, and tests whether the Galois group affects the constant.

Consequences for primes. Let $\mathbf{1}_t$ be the indicator that $f(t)$ is prime, $p_t=\min(1,C(f)/\log f(t))$ its Bateman–Horn probability, $A_T=\sum_{t<T}\mathbf{1}_t$ and $V_T=\sum_{t<T}p_t(1-p_t)$ the independent-trials variance. If the Hardy–Littlewood conjecture holds for the pair $(f(t),f(t+h))$, i.e. $\Pr[\mathbf{1}_t\mathbf{1}_{t+h}=1]\approx S_f(h)\,w_tw_{t+h}$ with $w_t=1/\log f(t)$, then

$$\mathrm{Var}_{\mathrm{HL}}(A_T)=V_T+2\sum_{h=1}^{T-1}\bigl(S_f(h)-C(f)^2\bigr)\,W_T(h),\qquad W_T(h)=\sum_{t<T-h}w_tw_{t+h},\tag{5}$$

and inserting (3) gives

$$\frac{\mathrm{Var}}{\mathbb{E}}\approx 1-\frac{\log T}{\log X}+\frac{1+2\bar{A}_f(T)/C(f)}{\log X},\qquad X\approx f(T),\tag{6}$$

where $\bar{A}_f(T)$ is the average over $H<T$ of $\Sigma_f(H)+\frac{1}{2}C\log H$ (it tends to A_f). So the Montgomery–Soundararajan law (4) holds asymptotically for every polynomial, with a correction that depends on f only through $A_f/C(f)$; but $\Sigma_f(H)$ approaches its asymptotic form slowly, $\bar{A}_f(T)/C$ is of order -1 to -2 for $T\leq 10^3$, and the correction is a negative 0.1–0.2 at the lengths one can compute. Section 5 compares this with primes along 67 cells of polynomial sequences, with a random-integer control.

What is conditional on what. Three statements of very different status appear in this paper, and we separate them. (A) The Bateman–Horn conjecture (1) for a non-linear f, i.e. that f takes infinitely many prime values at the predicted density. It is not known for a single non-linear polynomial; for t^2+1 it is one of Landau's problems of 1912, and the obstruction is the parity barrier of sieve theory. No theorem of this paper depends on it. (B) The Hardy–Littlewood pair conjecture for f, which is stronger than (A). It is used only in (5) and Section 5, to translate Conjecture 1 into a statement about the variance of prime counts; the paper of Montgomery and Soundararajan is conditional in exactly the same way. (C) Conjecture 1 itself, and the Hypothesis (E) to which Theorem 6 reduces it. These are statements about the explicit arithmetic function S_f, provable or refutable with no conjecture about primes anywhere; Hypothesis (E) is a concrete equidistribution statement about the roots of f modulo d on average over d, in the family of the theorems of Hooley [13] and of Duke, Friedlander and Iwaniec [5] on the roots of quadratic congruences. The theorems of Section 3 are unconditional; Proposition 10 assumes the Riemann hypothesis for ζ and $L(s,\chi_D)$ and a contour argument that we have sketched. Section 6 records where our polynomial families come from (a d-dimensional Ulam spiral whose lattice lines carry polynomials of degree exactly d, Theorem 12). Section 7 discusses what is proved, what is heuristic, and what should be done next.

Reader's guide. Section 2 defines S_f and records its elementary properties. Section 3 contains the theorems: the exact identity (§3.1), the Dedekind zeta function (§3.2), the Artin L-function and the quadratic closed formula (§3.3), the Cesàro theorem and the reduction to the off-diagonal (§3.4), the numerical tests (§3.5) and the function-field identity (§3.6). Section 4 explains where the zeros of $L(s, \chi_D)$ enter and why they are hard to see numerically. Section 5 compares the prediction with actual prime counts, Section 6 records where our polynomial families come from, and Section 7 summarises what is proved, what is conjectured and what should be done next. A reader interested only in the main theorem needs Sections 2, 3.1, 3.2 and 3.4.

2 The pair singular series

We record the elementary properties of S_f that are used below. Write $R(h)$ for the resultant of $f(t)$ and $f(t+h)$ (up to the denominators of f), a nonzero integer for $h \neq 0$ since f is irreducible and not constant. For $p \nmid R(h)$ the polynomials $f(t)$ and $f(t+h)$ have no common root modulo p, so $\nu_f(p,h) = 0$ and the local factor of (2) equals $(1 - 2\omega_f(p)/p)/(1 - 1/p)^2 = (1-\omega_f(p)/p)^2(1-1/p)^{-2} \cdot P_p$ with $P_p = (1-2\omega_f(p)/p)/(1-\omega_f(p)/p)^2 = 1-\omega_f(p)^2/(p-\omega_f(p))^2$; since $\sum_p \omega_f(p)^2/p^2 < \infty$, the product converges absolutely and

$$S_f(h) = C(f)^2 \prod_p P_p \prod_{p|R(h)} \frac{1+g_p(h)}{P_p}, \qquad 1 + g_p(h) = \frac{1 - \omega_{f,2}(p,h)/p}{(1 - \omega_f(p)/p)^2},$$

a finite correction to the constant $C(f)^2 \prod_p P_p$. In particular $S_f(h) \geq 0$, with $S_f(h) = 0$ exactly when some prime divides $f(t)f(t+h)$ for every t. For $f(t) = t$ one has $\omega_f = 1$, $\nu_f(p,h) = \mathbf{1}_{p|h}$, and $S_f = \mathfrak{S}$ is the twin-prime singular series $2\prod_{p>2}(1-(p-1)^{-2}) \prod_{p|h,p>2} \frac{p-1}{p-2}$ for even h. For a general f the correction factors depend on h through the residues of the differences of roots of f modulo p, and Section 3 makes this dependence exact. The pair-correlation interpretation of S_f, and the prime counts it predicts, are taken up in Section 5. The Bateman–Horn conjecture is open for every non-linear f, but Kravitz, Woo and Xu [16] have shown that a polynomial analogue of the prime-tuples conjecture holds for 100% of polynomials in an L^2 sense, and Banks and Ford [2] that random sets of the right density satisfy Bateman–Horn statistics almost surely.

3 The sum of the pair singular series

This section contains the theorems of the paper. The idea is simple: expand $S_f(h)$ into a sum over moduli d of terms that depend on h only through $h \bmod d$, sum over $h \leq H$, and read off the secondary term from the moduli whose contribution does not average out. The pairs of equal roots of f modulo d give a Dirichlet series with a pole whose residue is $1/C(f)$; the pairs of distinct roots are the obstacle.

Throughout this section $f \in \mathbb{Z}[t]$ is irreducible of degree $n \geq 1$, with positive leading coefficient and no fixed prime divisor, so that $\omega_f(p) < p$ for every p; $R(h)$ denotes the resultant of $f(t)$ and $f(t+h)$, a non-zero integer for $h \neq 0$ (two roots of an irreducible polynomial never differ by a non-zero integer), and $K = \mathbb{Q}[t]/(f)$. (The spiral polynomials of Section 6 have denominators dividing $d!$; everything below goes through with the period $p^{v_p(\mathrm{den})+1}$ in place of p at the finitely many primes dividing the denominator, as implemented in the code.)

3.1 An exact identity

For squarefree q let $\rho_q(a) = \sum_{s \bmod q,\ f(s)\equiv 0} e(as/q)$ be the exponential sum over the roots of f modulo q, and define the *root Ramanujan sum*

$$c_q^f(h) = \sum_{\substack{a \bmod q \\ (a,q)=1}} |\rho_q(a)|^2\, e(-ah/q) = \sum_{s,s' \bmod q} c_q(s'-s-h),$$

where c_q is the classical Ramanujan sum; c_q^f is multiplicative in q by the Chinese remainder theorem, and $c_p^f(h) = p\,\nu_f(p,h) - \omega_f(p)^2$. For $f(t)=t$ one has $c_q^f = c_q$.

Theorem 2. *Let $b(q) = \prod_{p|q}(p-\omega_f(p))^{-2}$. For every $h \geq 1$,*

$$\frac{S_f(h)}{C(f)^2} = \sum_{q \text{ squarefree}} b(q)\, c_q^f(h), \tag{7}$$

the series converging absolutely. Consequently, for every $H \geq 1$,

$$\sum_{h\leq H}\big(S_f(h) - C(f)^2\big) = C(f)^2 \sum_{d\geq 2} W_f(d)\,\Psi_d(H), \qquad W_f(d) = d\,b(d)\prod_{p\nmid d}\Big(1 - \frac{\omega_f(p)^2}{(p-\omega_f(p))^2}\Big), \tag{8}$$

where d runs over squarefree integers, $\Psi_d(H) = \sum_{s,s'} \psi_d(s'-s,H)$, the sum over pairs of roots of f modulo d, with $\psi_d(m,H) = \#\{h \leq H :\ h \equiv m\ (d)\} - H/d$, and the series converges absolutely. Splitting the pairs (s,s') into $s=s'$ and $s\neq s'$,

$$\sum_{h\leq H}\big(S_f(h) - C^2\big) = -C^2 \sum_{d\geq 2} a_f(d)\Big\{\frac{H}{d}\Big\} + C^2\,\mathrm{Off}_f(H), \qquad a_f(d) = W_f(d)\,\omega_f(d), \tag{9}$$

with $\mathrm{Off}_f(H) = \sum_d W_f(d) \sum_{s\neq s'} \psi_d(s'-s,H)$.

Proof. Put $a_p = \omega_f(p)/p$. By (2) and (1), $S_f(h)/C^2 = \prod_p(1+g_p(h))$ with $g_p(h) = (\nu_f(p,h)/p - a_p^2)/(1-a_p)^2 = c_p^f(h)/(p-\omega_f(p))^2$. For $p \nmid R(h)$ one has $\nu_f(p,h)=0$, so $g_p(h) = -a_p^2/(1-a_p)^2 = O(p^{-2})$ and $\sum_p |g_p(h)| < \infty$; hence the product converges absolutely and equals $\sum_q \prod_{p|q} g_p(h)$, which is (7) by multiplicativity. Summing over $h \leq H$ (a finite sum) and using $\sum_{h\leq H} c_q(m-h) = \sum_{d|q} d\,\mu(q/d)\,\#\{h \leq H :\ d \mid m-h\}$, we get for $q>1$

$$\sum_{h\leq H} c_q^f(h) = \sum_{d|q} d\,\mu(q/d) \sum_{s,s' \bmod q}\Big(\frac{H}{d} + \psi_d(s'-s,H)\Big) = \sum_{d|q,\ d>1} d\,\mu(q/d)\,\omega_f(q/d)^2\,\Psi_d(H),$$

because $\sum_{d|q}\mu(q/d)=0$ and pairs of roots modulo q project onto pairs modulo d with multiplicity $\omega_f(q/d)^2$. Writing $q=dm$ with $(m,d)=1$ and summing over m, $\sum_{(m,d)=1}\mu(m)b(m)\omega_f(m)^2 = \prod_{p\nmid d}(1-\omega_f(p)^2/(p-\omega_f(p))^2)$, an absolutely convergent product; this gives (8). The rearrangement is justified because $\sum_d d\,b(d)|\Psi_d(H)| < \infty$: for fixed h, $\nu_f(d,h)\neq 0$ forces $d \mid R(h)$ up to a bounded factor, so $\sum_d d\,b(d)\sum_{h\leq H}\nu_f(d,h)$ is a finite sum, while $\sum_d d\,b(d)\,\omega_f(d)^2 H/d < \infty$. Finally $\psi_d(0,H) = -\{H/d\}$ gives (9). □

The diagonal part of (9) is a sum of fractional parts weighted by the multiplicative function a_f; the off-diagonal part collects, for every d, the discrepancies of the residue classes $s'-s$ of *differences of distinct roots* of f modulo d, each of size $O(1)$ and of mean zero over the residues. For linear f there are no distinct roots and $\mathrm{Off}_f \equiv 0$.

3.2 The diagonal: the Dedekind zeta function and the residue $1/C(f)$

Theorem 3. *The Dirichlet series $D_f(s) = \sum_d a_f(d) d^{-s}$ has the Euler product $D_f(s) = \prod_p \big(P_p + p\, b(p)\, \omega_f(p)\, p^{-s}\big)$ with $P_p = 1 - \omega_f(p)^2/(p - \omega_f(p))^2$, and*

$$D_f(s) = \zeta_K(s+1)\, E_f(s)$$

with E_f holomorphic in $\Re s > -\frac{1}{2}$. In particular D_f is meromorphic there with a single simple pole at $s = 0$, and

$$\operatorname*{Res}_{s=0} D_f(s) = \prod_p \frac{1 - 1/p}{1 - \omega_f(p)/p} = \frac{1}{C(f)}, \qquad \sum_{d \le x} a_f(d) = \frac{\log x}{C(f)} + B_f + O\Big(\frac{1}{\log x}\Big).$$

Proof. a_f is supported on squarefree d with $a_f(d) = \prod_{p|d} p\, b(p) \omega_f(p) \prod_{p\nmid d} P_p$, which gives the Euler product. At $s = 0$ the local factor is $P_p + p\, b(p) \omega_f(p) = \big((p-\omega)^2 - \omega^2 + p\omega\big)/(p-\omega)^2 = p/(p - \omega_f(p))$. For large p, $P_p + p\, b(p)\omega_f(p) p^{-s} = 1 + \omega_f(p) p^{-1-s} + O(p^{-2})$ uniformly in $\Re s \ge -\frac{1}{2} + \delta$, while for every p not dividing the discriminant of f the Euler factor of $\zeta_K(s+1)$ is $1 + \omega_f(p) p^{-1-s} + O(p^{-2-2\Re s})$ by the Dedekind–Kummer theorem ($\omega_f(p)$ is then the number of degree-one prime ideals above p). The quotient is therefore $1 + O(p^{-2+2\delta})$ and E_f is an absolutely convergent Euler product, holomorphic in $\Re s > -\frac{1}{2}$. Since ζ_K has a simple pole at 1, so has D_f at 0, and $\operatorname{Res}_{s=0} D_f = \lim_{s\to 0^+} s D_f(s) = \lim_{s \to 0^+} \prod_p (P_p + p b(p) \omega_f(p) p^{-s})(1 - p^{-1-s})$; the logarithm of the product is $\sum_p (\omega_f(p) - 1) p^{-1-s} +$ (absolutely convergent), and $\sum_p (\omega_f(p) - 1) p^{-1-s}$ converges at $s = 0$ by the prime ideal theorem ($\sum_{p \le x} (\omega_f(p) - 1) = o(x/\log x)$), hence uniformly for $s \ge 0$, so the limit is the value at $s = 0$, $\prod_p \frac{p}{p - \omega_f(p)} (1 - \frac{1}{p}) = 1/C(f)$. The asymptotic for $\sum_{d \le x} a_f(d)$ follows from the Selberg–Delange method [21, II.5] applied to $\sum_d d\, a_f(d)\, d^{-s} = D_f(s-1) = \zeta(s) \cdot \frac{\zeta_K(s)}{\zeta(s)} E_f(s-1)$, whose second factor is holomorphic and non-vanishing in a neighbourhood of $s = 1$ and in a standard zero-free region, followed by partial summation. ☐

Theorem 3 is the rigorous content of the phrase "the correlation carried by a prime p dividing h has weight $\omega_f(p)/(p - \omega_f(p))$": the partial sums of that weight grow like $\log H / C(f)$, exactly as $\sum_{k \le H} \mu(k)^2/\varphi(k) \sim \log H$ does in the Montgomery–Soundararajan computation [18] for $f(t) = t$. It also exhibits the Dedekind zeta function of the field cut out by f as the generating function of the diagonal.

3.3 The Artin L-function of the symmetric square, and quadratics in closed form

The factor E_f of Theorem 3 is not just "holomorphic near 0": it is itself an L-function, and this is what connects the second moment to zeros. Let V_f be the permutation representation of $G = \mathrm{Gal}(f)$ on the roots of f, and $\Lambda^k V_f$, $\mathrm{Sym}^2 V_f$ its exterior powers and symmetric square. For p not dividing the discriminant of f, $\omega_f(p)$ is the number of fixed points of Frob_p on the roots, and the Euler factor of $1/\zeta_K(w)$ at p is $\det(1 - \mathrm{Frob}_p\, p^{-w} \mid V_f) = \sum_k (-1)^k \chi_{\Lambda^k V_f}(\mathrm{Frob}_p) p^{-kw}$.

Theorem 4. *Write $\chi_2(p) = \chi_{\mathrm{Sym}^2 V_f}(\mathrm{Frob}_p) = \frac{1}{2}\omega_f(p)(\omega_f(p) + 1) + n_2(p)$, where $n_2(p)$ is the number of 2-cycles of Frob_p on the roots. Then*

$$E_f(s) = \frac{H_f(s)}{L(2s+2, \mathrm{Sym}^2 V_f)},$$

with $H_f(s)$ an Euler product whose factors are $1 + O(p^{-2}) + O(p^{-2-s}) + O(p^{-3-3s})$, absolutely convergent for $\Re s > -\frac{2}{3}$. For a quadratic f of discriminant D, with χ_D the Kronecker symbol,

$$D_f(s) = \frac{\zeta(s+1)\, L(s+1, \chi_D)}{\zeta(2s+2)^2\, L(2s+2, \chi_D)}\, H_f(s),$$

and extracting the cubic terms as well, $H_f(s) = \zeta(3s+3)L(3s+3,\chi_D)\,\tilde{H}_f(s)$ with $\tilde{H}_f$ absolutely convergent for $\Re s > -\frac{3}{4}$. The extraction does not stop: writing $v = p^{-(s+1)}$, the part of the local factor of E_f at a split prime that survives as $p \to \infty$ with v fixed is $(1+2v)(1-v)^2 = 1-3v^2+2v^3$, whose logarithm has infinitely many terms, so that $\tilde{H}_f(s) = \zeta(4s+4)^{-1}L(4s+4,\chi_D)^{-2}\,\zeta(5s+5)^3L(5s+5,\chi_D)^3\cdots M_f(s)$ with integer exponents and M_f absolutely convergent for $\Re s > -1$; D_f is meromorphic in $\Re s > -1$, has a real pole at $s = -\frac{2}{3}$ (from $\zeta(3s+3)$) and further real poles at $s = -1+1/k$, k odd, and $\Re s = -1$ is a natural boundary, the zeros of the local factors F_p accumulating at every point of it.

Proof. The local factor of D_f is $F_p(s) = P_p + p\,b(p)\omega_f(p)p^{-s}$; it contains p^{-s} only to the first power. Multiplying by $\det(1-\mathrm{Frob}_p\,p^{-1-s} \mid V_f) = 1-\omega p^{-1-s}+\chi_{\Lambda^2}p^{-2-2s}-\cdots$ and using $F_p = 1-\omega^2p^{-2}+\omega p^{-1-s}+2\omega^2p^{-2-s}+O(p^{-3})$, the coefficient of p^{-2-2s} in the E_f-factor is $\chi_{\Lambda^2}(\mathrm{Frob}_p)-\omega^2 = -\chi_{\mathrm{Sym}^2}(\mathrm{Frob}_p)$, since $V\otimes V = \Lambda^2V\oplus\mathrm{Sym}^2\,V$; and $\chi_{\mathrm{Sym}^2}(g) = \frac{1}{2}(\chi_V(g)^2+\chi_V(g^2))$ with $\chi_V(g^2) = \omega+2n_2$ gives the stated formula. The remaining terms are $-\omega^2p^{-2}$, $2\omega^2p^{-2-s}$, and terms $p^{-k(1+s)}$, $k\geq 3$, with bounded coefficients, whence the convergence statement. For a quadratic, $V_f = \mathbf{1}\oplus\chi_D$, so $\zeta_K = \zeta L(\chi_D)$, $\mathrm{Sym}^2\,V_f = \mathbf{1}\oplus\mathbf{1}\oplus\chi_D$, and the cubic coefficient is $\chi_{\Lambda^3}-\omega\chi_{\Lambda^2} = -\omega\chi_D(p)$ for $\dim V = 2$; with $\omega = 1+\chi_D(p)$ this is $-(\chi_D+1)$ up to the sign convention, giving the factor $\zeta(3s+3)L(3s+3,\chi_D)$. Continuing the expansion of $\log\big((1+2v)(1-v)^2\big)$ (split p) and $\log(1-v^2)$ (inert p) and inverting over the divisors of each degree gives the exponents of the higher factors ($-3v^4$ at split primes is the first uncancelled term, whence the abscissa $-\frac{3}{4}$); the mixed terms $p^{-a}v^b$, $a\geq 1$, converge absolutely for $\Re s > -1$. In the region $\Re s > -1+1/(K+1)$ the finitely extracted product vanishes exactly where a local factor $F_p(s) = P_p+p\,b(p)\omega_f(p)p^{-s}$ does, i.e. at $p^{-s} = -P_p/(p\,b(p)\omega_f(p))$; these zeros have real part $\to -1$ and imaginary parts $(2j+1)\pi/\log p$, dense as $p\to\infty$, which gives the natural boundary. The exponents and the convergence claims are checked to high precision in the code accompanying the sequel. □

The poles of $1/L(2s+2,\mathrm{Sym}^2\,V_f)$ lie at $s = \rho/2-1$ for the zeros ρ of the symmetric-square L-function; for a quadratic these are the zeros of ζ (double) and of $L(s,\chi_D)$ (simple). In the Perron integral for the Riesz means of the diagonal part they produce terms $x^{\rho/2}$, exactly as the zeros of ζ do through $1/\zeta(2s)$ in the work of Goldston and Suriajaya [11] on $\sum_{h\leq x}\mathfrak{S}(h)$; see Section 4.

For quadratics the pair singular series itself has a closed form of twin-prime type.

Theorem 5. *Let $f = at^2+bt+c\in\mathbb{Z}[t]$ be irreducible with discriminant $D = b^2-4ac$, and let $Q(h) = a^2h^2-D$. For every prime $p\nmid 2aD$: $\omega_f(p) = 1+\chi_D(p)$, the two roots of f modulo a split prime differ by $\pm\sqrt{D}/a$, and hence $\nu_f(p,h) = \omega_f(p)$ if $p\mid h$, $\nu_f(p,h) = 1$ if $p\mid Q(h)$ and $p\nmid h$, and $\nu_f(p,h) = 0$ otherwise. Consequently*

$$S_f(h) = C(f)^2\prod_p P_p\prod_{p\mid h}\frac{p-\omega_f(p)}{p-2\omega_f(p)}\prod_{\substack{p\mid Q(h)\\ p\nmid h}}\frac{p-3}{p-4},$$

with the finitely many primes dividing $2aD$ carrying their exact local factors instead. For $f = t^2+1$: $S_f(h) = 0$ for odd h and, for even h, $S_f(h) = 2C(f)^2\prod_{p\equiv 1(4)}P_p\prod_{p\mid h,\,p\equiv 1(4)}\frac{p-2}{p-4}\prod_{p\mid h^2+4,\,p\nmid h}\frac{p-3}{p-4}$.

Proof. For $p\nmid 2aD$ split, the roots are $(-b\pm\sqrt{D})/2a$, so the ordered pairs of distinct roots have differences $\pm\sqrt{D}/a$, and $h\equiv\pm\sqrt{D}/a$ iff $p\mid a^2h^2-D$; for p odd exactly one of the two signs occurs, so $\nu_f(p,h) = 1$. Then $1+g_p(h)$ equals $1/(1-\omega/p)$ for $p\mid h$, $(1-3/p)/(1-2/p)^2$ for $p\mid Q(h)$, $p\nmid h$ (where $\omega = 2$), and P_p otherwise; dividing by P_p gives the factors $(p-\omega)/(p-2\omega)$ and $(p-3)/(p-4)$. For t^2+1, $\omega(2) = 1$ makes $P_2 = 0$ and the local factor at 2 equal to 2 for even h and 0 for odd h. □

The factor $(p-\omega)/(p-2\omega)$ for $p \mid h$ is the analogue of the twin-prime factor $(p-1)/(p-2)$; the new ingredient is the second product over the primes represented by the quadratic form $Q(h)$. This makes $S_f(h)$ computable exactly, by sieving h and $Q(h)$, for all h up to $4 \cdot 10^7$ in a minute.

3.4 Theorems

The fractional-part sum $\sum_d a_f(d)\{H/d\}$ equals $\frac{1}{2C}\log H + O(1)$ if the fractional parts $\{H/d\}$ have mean $\frac{1}{2}$ against the weights $a_f(d)$; for $f(t) = t$ this is the theorem of Friedlander–Goldston and Vaughan [7, 23] ($\sum_{h\le H}\mathfrak{S}(h) = H - \frac{1}{2}\log H + O((\log H)^{2/3})$), whose proof needs exponential-sum estimates of Vinogradov type. Under a Cesàro weight the fractional parts are tamed and an elementary argument suffices.

Theorem 6. *Let $\phi(\theta) = \theta - \theta^2$. For $H \ge 2$,*

$$\sum_{h\le H}\Big(1-\frac{h}{H}\Big)\big(S_f(h) - C(f)^2\big) = -\tfrac{1}{2}\,C(f)\log H + O_f(1) + \frac{C(f)^2}{H}\operatorname{Off}^*_f(H), \tag{10}$$

where, with $m = s' - s \bmod d \in [1, d-1]$ and the sum over unordered pairs of distinct roots,

$$\operatorname{Off}^*_f(H) = \sum_{d\ge 2} W_f(d)\sum_{\{s,s'\}}\Big[-\frac{m(d-m)}{d} + \frac{d}{2}\Big(\phi\Big(\Big\{\frac{H-m}{d}\Big\}\Big) + \phi\Big(\Big\{\frac{H+m}{d}\Big\}\Big)\Big)\Big]$$

converges absolutely. In particular:

- *(i) if $\omega_f(p) \le 1$ for all p, e.g. for every linear f, then $\operatorname{Off}^*_f \equiv 0$ and $\sum_{h\le H}(1-h/H)(S_f(h) - C^2) = -\frac{1}{2}C\log H + O_f(1)$;*
- *(ii) for general f, Conjecture 1 in Cesàro form holds, with the leading term $-\frac{1}{2}C(f)\log H$, if and only if $\operatorname{Off}^*_f(H) = o(H\log H)$; trivially $\operatorname{Off}^*_f(H) \ll H(\log H)^{r_f - 1}$, where $r_f \ge 2$ is the number of orbits of the Galois group of f on ordered pairs of roots.*

Proof. Multiply (7) by $H - h$ and sum. As in the proof of Theorem 2,

$$\sum_{h\le H}(H-h)\big(S_f(h) - C^2\big) = C^2\sum_{d\ge 2}W_f(d)\sum_{s,s' \bmod d}B_d(s'-s), \qquad B_d(m) = \sum_{\substack{h\le H\\ h\equiv m\ (d)}}(H-h) - \frac{H^2}{2d},$$

the subtraction of $H^2/(2d)$ being allowed because $\sum_{d\mid q}\mu(q/d) = 0$ for $q > 1$. An elementary computation gives, for $1 \le m \le d$ and every $d \ge 1$ (also for $d > H$),

$$B_d(m) = -\frac{Hm}{d} + \frac{m^2}{2d} + \frac{H}{2} - \frac{m}{2} + \frac{d}{2}\phi\Big(\Big\{\frac{H-m}{d}\Big\}\Big), \qquad \text{so} \qquad B_d(d) = -\frac{H}{2} + \frac{d}{2}\phi\Big(\Big\{\frac{H}{d}\Big\}\Big),$$

and $B_d(m) + B_d(d-m)$ is the bracket in the definition of Off^*_f. The diagonal is

$$\sum_d a_f(d)B_d(d) = -\frac{H}{2}\sum_{d\le H}a_f(d) + \frac{1}{2}\sum_{d\le H}d\,a_f(d)\,\phi\Big(\Big\{\frac{H}{d}\Big\}\Big) - \frac{H^2}{2}\sum_{d>H}\frac{a_f(d)}{d},$$

using $B_d(d) = -H^2/(2d)$ for $d > H$. By Theorem 3 the first term is $-\frac{H}{2C}\log H + O(H)$ and, by partial summation, $\sum_{d>H}a_f(d)/d = 1/(CH) + o(1/H)$, so the third term is $O(H)$. In the second, $0 \le \phi \le \frac{1}{4}$ and $\sum_{d\le H}d\,|a_f(d)| = O(H)$ by Shiu's theorem [19] for the non-negative multiplicative function $d\,|a_f(d)|$, whose value at p is $\omega_f(p)(1 + O(1/p))$ and satisfies $\sum_{p\le x}(\omega_f(p) - 1)/p = O(1)$. Dividing by H gives (10). Absolute convergence of Off^*_f and the bound in (ii) follow from $|B_d(m)| \ll d + H^2/d$ for $d > H$, $\ll d$ for $d \le H$, together with $\sum_{d\le x}|W_f(d)|\,\omega_f(d)^2\,d \ll x(\log x)^{r_f - 1}$ (Shiu's theorem again, since the mean value of $\omega_f(p)^2$ over primes is r_f by Chebotarev) and the finiteness of $\sum_d W_f(d)\sum_{h\le H}\nu_f(d,h)(H-h)$. If $\omega_f(p) \le 1$ for all p there are no pairs of distinct roots and $\operatorname{Off}^*_f = 0$. $\square$

For $f(t)=t$, Theorem 6(i) is the Cesàro form of the Friedlander–Goldston–Vaughan asymptotic, with an elementary proof of the leading term; Goldston and Suriajaya [11, 10] have shown that the error term of the Riesz means of $\sum_{h\le x}\mathfrak{S}(h)$ is an explicit formula over the zeros of ζ. In our setting the generating series is $\zeta_K(s+1)E_f(s)$, and the same analysis would express the fluctuations of $\sum_{h\le H}S_f(h)$ through the zeros of the Dedekind zeta function of K; we return to this in Section 7.

Remark 7. Part (ii) deserves emphasis. The Galois group of f acts on ordered pairs of distinct roots; it is 2-transitive exactly when $r_f=2$, which is the generic case (S_n, A_n, and every 2-transitive group). Then the trivial bound is $\mathrm{Off}^*_f(H)\ll H\log H$, and the leading term $-\frac{1}{2}C(f)\log H$ of Conjecture 1 in Cesàro form follows from *any* saving of an unbounded factor over the trivial bound: one needs the residue classes $s'-s \bmod d$ of differences of distinct roots to be distributed among the $d-1$ nonzero classes with a discrepancy that is $o(1)$ on average over $d\le H$ against the weights $W_f(d)$, not a power saving. For a quadratic the pairs are $(\pm\sqrt{D})$ and the differences $\pm 2\sqrt{D} \bmod d$; the equidistribution of the roots of $x^2\equiv D \pmod d$ on average over d, with a power saving, is Hooley's theorem [13], and the analogous statements for prime moduli are those of Duke, Friedlander and Iwaniec [5] and Tóth [22]. What is not in the literature is the average with the multiplicative weights $W_f(d)$ and the fractional-part test function; we set out the transcription in a companion note and do not claim it here. For $r_f\ge 3$ the trivial bound has an extra power of $\log H$ and a genuine cancellation among the orbits is required (Section 4.3).

The full conjecture (3), with a constant term, needs more than the leading term:

Hypothesis (E). $\mathrm{Off}_f(H)=O(1)$, equivalently $\mathrm{Off}^*_f(H)=O(H)$: the fractional parts $\{(H\pm m)/d\}$, m ranging over the differences of distinct roots of f modulo d, have mean $\frac{1}{2}$ against the weights $W_f(d)$.

This is the equidistribution statement of Remark 7 with a bounded, rather than merely $o(H\log H)$, error; we have not proved it. We test it directly for quadratics, where $\omega_f(p)=1+\left(\frac{\operatorname{disc} f}{p}\right)$ allows the diagonal sum $\sum_d a_f(d)\{H/d\}$ to be computed exactly to $d\le 4{,}000{,}000$, so that $\mathrm{Off}_f(H)$ is obtained from (9) by subtraction (Section 3.5).

3.5 Numerical verification

We computed $\Sigma_f(H)=\sum_{h\le H}(S_f(h)-C(f)^2)$ for $H\le 10^5$, with every Euler product taken over $p\le 60{,}000$ using the exact period $p^{v_p(\mathrm{den})+1}$, for 110 polynomials: 8 progressions $a+qt$ with $q\le 300$; sixteen "spiral" polynomials each of degree $2,3,4$ (eventual ray polynomials of the spiral of Section 6, rational coefficients with denominator dividing $d!$, leading coefficient 2^d); sixteen random integer polynomials each of degree $2,3,4$ (coefficients in $[-60,60]$, leading coefficient in $[1,8]$); and six named polynomials including Euler's t^2+t+41, t^2+1 and t^3+2. Polynomials with a fixed divisor or a square discriminant were excluded. The script `thesis/sumS.ts` reproduces everything; no primes are used.

Table 1 and Figure 1 show the result. Over the last computed decade the slope divided by C is -0.53 ± 0.05 for the progressions and -0.50 ± 0.27 (standard error 0.03) for the 102 non-linear polynomials, with no visible dependence on the degree. The scatter of individual slopes is not statistical error but genuine arithmetic fluctuation of $\Sigma_f(H)$ (the terms $S_f(h)-C^2$ are large and of both signs, and the partial sums over one decade of H do not average them completely); it is largest for polynomials with large C, which have no roots modulo the small primes and hence the most irregular S_f. The constant A_f/C is negative and of order -1 for most polynomials, i.e. the sum is more negative at finite H than $-\frac{1}{2}C\log H$; this is the quantity that governs the finite-T variance in (6). Because the mean C of this sample is close to 1, Table 1 cannot by itself separate C from C^2; that is the purpose of the next test.

Table 1: Growth of $\Sigma_f(H) = \sum_{h \le H}(S_f(h) - C^2)$. Slopes are increments per unit of $\log H$ over the indicated decade, divided by C or by C^2 (mean $\pm$ s.d. over the polynomials of the group). A_f/C is the constant term of (3) estimated at $H = 10^5$. Conjecture 1 predicts -0.50 in the "$/C$" columns for every group.

group	d	n	mean C	slope/C, 10^3–10^4	slope/C, 10^4–10^5	slope/C^2, 10^4–10^5	A_f/C
arith. progressions	1	8	1.49	-0.40 ± 0.07	-0.53 ± 0.05	-0.41 ± 0.14	-0.31 ± 0.30
quadratics (spiral)	2	16	1.61	-0.27 ± 0.43	-0.65 ± 0.40	-0.56 ± 0.64	-0.49 ± 1.14
quadratics (random)	2	16	1.29	-0.44 ± 0.25	-0.48 ± 0.20	-0.56 ± 0.60	-0.51 ± 1.03
cubics (spiral)	3	16	1.57	-0.49 ± 0.20	-0.51 ± 0.20	-0.38 ± 0.22	-0.36 ± 0.48
cubics (random)	3	16	1.25	-0.55 ± 0.23	-0.37 ± 0.27	-0.42 ± 0.46	-0.08 ± 0.73
quartics (spiral)	4	16	1.38	-0.35 ± 0.23	-0.50 ± 0.18	-0.56 ± 0.49	-0.42 ± 0.70
quartics (random)	4	16	1.62	-0.38 ± 0.18	-0.51 ± 0.17	-0.50 ± 0.41	-0.30 ± 0.68

Table 2: The exact test for 25 quadratics ($H \le 10^6$; `thesis/exact.ts`). $\Sigma_f(H)$ is computed from Theorem 5 by sieving h and $Q(h) = a^2h^2 - D$; $C(f)$ from $L(1, \chi_{D_0})$ and an absolutely convergent correction. Slope is the increment per unit of $\log H$ between $H = 10^3$ and 10^6. Conjecture 1 predicts -0.50 asymptotically in the "$/C$" column for every f; a C^2 law would put a constant in the "$/C^2$" column.

f	D	$C(f)$	$\Sigma_f(10^3)$	$\Sigma_f(10^6)$	slope/C	slope/C^2
$7t^2 - 4t + 39$	-1076	0.390	-1.50	-2.74	-0.459	-1.177
$7t^2 - 6t + 35$	-944	0.450	-2.50	-4.11	-0.519	-1.152
$7t^2 - 4t - 18$	520	0.524	-2.75	-4.44	-0.467	-0.892
$5t^2 - 6t - 17$	376	0.607	-3.31	-5.10	-0.427	-0.704
$7t^2 - 34$	952	0.699	-3.77	-5.81	-0.423	-0.605
$3t^2 - 4t + 38$	-440	0.810	-3.38	-6.28	-0.518	-0.639
$2t^2 - 4t + 33$	-248	0.940	-4.43	-5.62	-0.183	-0.194
$5t^2 + 6$	-120	1.087	-4.73	-8.65	-0.522	-0.480
$t^2 - 6t + 24$	-60	1.267	-4.79	-9.85	-0.579	-0.457
$t^2 + 1$	-4	1.373	-4.53	-8.05	-0.371	-0.270
$2t^2 + 1$	-8	1.426	-5.84	-11.14	-0.538	-0.377
$2t^2 - 5t + 16$	-103	1.463	-4.47	-11.28	-0.674	-0.461
$7t^2 - t + 7$	-195	1.706	-7.75	-11.54	-0.321	-0.188
$t^2 - 2$	8	1.850	-6.25	-11.38	-0.401	-0.217
$t^2 - 6t + 16$	-28	1.973	-7.04	-13.62	-0.483	-0.245
$3t^2 - 5t - 9$	133	2.298	-11.19	-13.76	-0.162	-0.070
$5t^2 + 26$	-520	2.642	-7.57	-14.28	-0.368	-0.139
$3t^2 - 3t + 31$	-363	3.057	-15.09	-19.34	-0.201	-0.066
$t^2 + t + 11$	-43	3.259	-14.73	-19.75	-0.223	-0.068
$t^2 - 5t - 25$	125	3.547	-12.67	-23.03	-0.423	-0.119
$t^2 + t + 17$	-67	4.175	-16.16	-27.74	-0.401	-0.096
$t^2 - 5t + 23$	-67	4.175	-16.16	-27.74	-0.401	-0.096
$7t^2 - 3t + 37$	-1027	4.716	-18.36	-28.22	-0.303	-0.064
$3t^2 - 3t + 23$	-267	5.637	-18.31	-32.99	-0.377	-0.067
$t^2 + t + 41$	-163	6.640	-17.49	-39.21	-0.473	-0.071

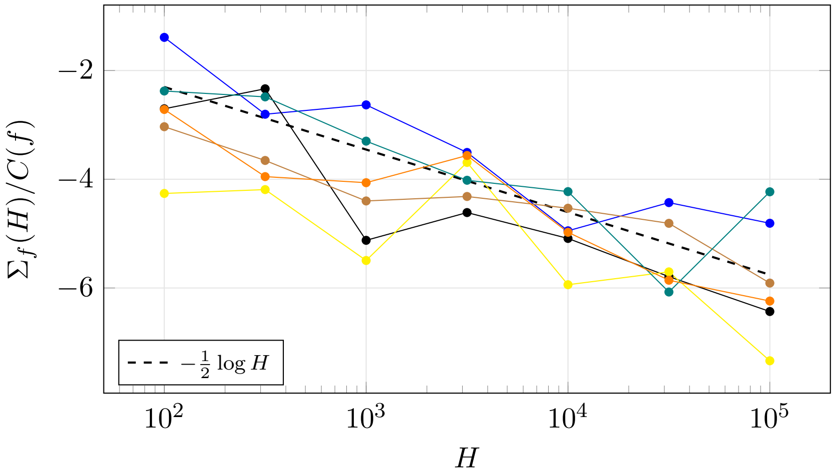

Figure 1: $\Sigma_f(H)/C(f)$ for the six named polynomials (t^2+t+41, $4t^2-2t+1$, $4t^2+3t+17$, $4t^2+30t+7$, t^2+1, t^3+2) against the reference slope $-\frac{1}{2}\log H$. The curves are parallel to the reference up to arithmetic fluctuations, with polynomial-dependent offsets A_f/C; the fluctuations are largest for t^2+t+41, whose large C (no roots modulo any $p < 41$) makes $S_f(h)/C^2$ vary most strongly with h.

The exact test: C against C^2. Theorem 5 removes the Euler-product truncation altogether for quadratics: $S_f(h)/C^2$ is $\prod_p P_p$ (an absolutely convergent product, terms $1+O(p^{-2})$) times a finite product over the primes dividing h and $Q(h) = a^2h^2 - D$, which we obtain by sieving $h \le 10^6$ and $Q(h)$ with the roots $\pm\sqrt{D}/a$ of Q modulo each prime up to $\sqrt{Q(10^6)}$; the remaining cofactor is 1 or a prime. $C(f)$ is computed from $L(1, \chi_{D_0})$, D_0 the fundamental discriminant, by the digamma formula, times an absolutely convergent correction. We checked the implementation against the definition (2) for four quadratics and $h \le 3000$, truncating both at the same set of primes: the two agree to $1 \cdot 10^{-14}$ in relative terms, i.e. to rounding, including the vanishing values. We then chose 25 irreducible quadratics without fixed divisor, $a \le 7$, with $C(f)$ spread uniformly in $\log C$ over $[0.39, 6.64]$, a lever arm of more than a decade in C, and fitted

$$\frac{\Sigma_f(10^6) - \Sigma_f(10^3)}{\log 10^3} = -k\, C(f)^\alpha$$

in two ways (Table 2, Figure 2): by least squares on $\log(-\text{slope})$ against $\log C$, which weights every polynomial equally in relative terms, and by unweighted least squares on the slopes themselves, which weights the large-C polynomials more. The results are

$$\alpha = 0.87 \pm 0.09,\ k = 0.41 \quad \text{(logarithmic)}; \qquad \alpha = 1.10 \pm 0.11,\ k = 0.33 \pm 0.06 \quad \text{(unweighted)},$$

respectively 1.5 and 0.9 standard errors from $\alpha = 1$, and 12.7 and 8.0 from $\alpha = 2$; the correlation of slope$/C$ with C is 0.26, that of slope$/C^2$ with C is 0.72. The two fits bracket $\alpha = 1$ and both exclude $\alpha = 2$: the first power of C is the law. The mean slope$/C$ is -0.41 ± 0.12 (standard error 0.02), i.e. k is about 0.82 of the asymptotic $\frac{1}{2}$; by decades the mean slope$/C$ is -0.34, -0.57, -0.31 over 10^3–10^4, 10^4–10^5, 10^5–10^6. The finite-H slope is thus systematically short of $\frac{1}{2}$ over three decades. This is what the exact identity predicts: the diagonal $-C^2\sum_d a_f(d)\{H/d\}$ approaches $\frac{1}{2C}\log H$ only as the mean of $\{H/d\}$ against $a_f(d)$ settles to $\frac{1}{2}$, and the drift of that mean over $d \le H$, together with the bounded oscillation of Off_f, is the slowly varying A_f of (3). The individual slopes scatter by ± 0.12 around the mean; this scatter, not the standard error of the fit, is the honest uncertainty of the coefficient at $H = 10^6$, and the value $\frac{1}{2}$ is a prediction of Theorem 3, not a measurement.

The off-diagonal remainder is bounded (Hypothesis (E) for quadratics). For 25 quadratics (the named ones and random ones with coefficients in $[-60, 60]$) we computed both

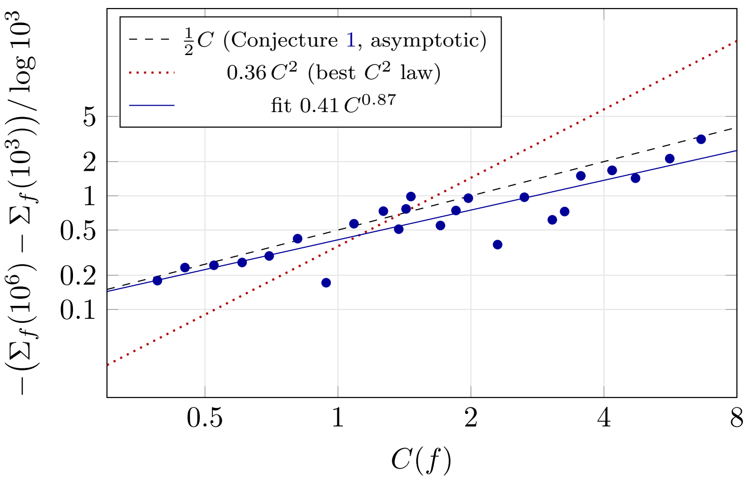

Figure 2: The exact test. For 25 quadratics the slope of $\Sigma_f(H)$ per unit of $\log H$ over $10^3 \leq H \leq 10^6$, computed from the closed formula of Theorem 5 with no Euler-product truncation, against $C(f)$ on logarithmic axes. A law $-kC^\alpha$ is a straight line of slope α; the logarithmic fit gives $\alpha = 0.87 \pm 0.09$ (unweighted: 1.10 ± 0.11). The dashed line is the conjectured asymptotic $\frac{1}{2}C$; the finite-H points lie below it by the drift of the constant term (Table 2).

sides of (9): the left side from the pair series ($p \leq 60{,}000$), the diagonal sum $\sum_d a_f(d)\{H/d\}$ exactly over squarefree $d \leq 4{,}000{,}000$ with $\omega_f(p)$ from Legendre symbols (the tail is about $H/(C\cdot 4{,}000{,}000)$), and $\mathrm{Off}_f(H)$ as the difference (Table 3; script `thesis/offdiag.ts`). Between $H = 10^3$ and $H = 10^5$ the two sides grow by 0.83 and 0.91 times $\frac{1}{2C}\log H$ respectively (averages over the 25 polynomials), while $\mathrm{Off}_f(H)$ stays bounded: mean 0.90, standard deviation 1.36, largest absolute value 4.98 at $H = 10^5$, and its change over the two decades is 0.08 times $\frac{1}{2C}\log H$. The whole logarithmic growth of $\Sigma_f(H)$ is carried by the diagonal, as Theorem 3 predicts, and the off-diagonal remainder behaves like the bounded oscillating quantity that Hypothesis (E) asserts.

Table 3: Both sides of (9) for the named quadratics at $H = 10^3$ and $H = 10^5$: $\Sigma_f(H)/C^2$, the diagonal $-\sum_d a_f(d)\{H/d\}$, and their difference $\mathrm{Off}_f(H)$. Conjecture 1 predicts a growth of $\frac{1}{2C}\log H$ for the first two columns, i.e. $2.30/C$ per decade, and $O(1)$ for the last.

f	C	Σ/C^2, 10^3	Σ/C^2, 10^5	diag, 10^3	diag, 10^5	Off, 10^3	Off, 10^5
t^2+t+41	6.64	-0.40	-0.72	-0.43	-0.82	0.03	0.09
$4t^2-2t+1$	2.24	-2.29	-2.87	-2.16	-3.13	-0.13	0.25
$4t^2+3t+17$	0.62	-8.84	-11.80	-11.14	-13.13	2.30	1.33
$4t^2+30t+7$	3.23	-1.36	-1.83	-1.37	-2.01	0.01	0.18
t^2+1	1.37	-2.40	-3.08	-2.14	-3.40	-0.26	0.31
$2t^2+1$	1.43	-2.87	-4.23	-3.08	-4.84	0.21	0.61
t^2+t+1	2.24	-2.08	-3.06	-2.16	-3.13	0.08	0.06

3.6 The function-field analogue: the off-diagonal vanishes for small moduli

Over $\mathbb{A} = \mathbb{F}_q[u]$, q odd, every object above has a verbatim analogue: for $f \in \mathbb{A}[t]$ irreducible in t, separable and without fixed prime divisor, $|P| = q^{\deg P}$ replaces p, "$h \leq H$" becomes "$h \neq 0$, $\deg h < N$" ($q^N - 1$ elements) and $\log H$ becomes N; the products $C(f)$, b, P_P, W_f, a_f converge absolutely as before. The identity (8) holds with one decisive simplification.

Theorem 8. *Let $f \in \mathbb{A}[t]$ be as above, let $a_f(d) = W_f(d)\,\omega_f(d)$ with W_f as in Theorem 2 (products over the primes of $\mathbb{A}$), and put $P(1) = \prod_P P_P = W_f(1)$. For every $N \geq 1$,*

$$\sum_{\substack{h\neq 0\\ \deg h<N}} \Big(\frac{S_f(h)}{C(f)^2} - 1\Big) = -\sum_{1\leq \deg d\leq N} a_f(d) \;-\; q^N \sum_{\deg d>N} \frac{a_f(d)}{|d|} \;+\; \big(1-P(1)\big) \;+\; \mathrm{Off}_f(N),$$

$$\mathrm{Off}_f(N) = \sum_{\deg d>N} W_f(d) \sum_{\substack{s,s' \bmod d\\ s\neq s'}} \Big(\mathbf{1}_{0\leq \deg \tilde m<N} - q^{N-\deg d}\Big), \qquad \tilde m \equiv s'-s\ (d),\ \deg \tilde m < \deg d,$$

where $\mathrm{Off}_f(N)$ *runs over squarefree d of degree $> N$ only. In particular the pairs of* distinct *roots modulo d contribute nothing for* $\deg d \leq N$*: the off-diagonal part of (9) vanishes identically for all moduli of degree $\leq N$.*

Proof. The expansion $S_f(h)/C^2 = \sum_Q b(Q) c_Q^f(h)$ over squarefree monic Q is proved as in Theorem 2, and for $Q \neq 1$

$$\sum_{\substack{h\neq 0\\ \deg h<N}} c_Q^f(h) = \sum_{d|Q} |d|\,\mu(Q/d)\,\omega_f(Q/d)^2 \sum_{s,s' \bmod d} \mathrm{cnt}_d(s'-s),$$

where $\mathrm{cnt}_d(m) = \#\{h \neq 0 :\ \deg h < N,\ h \equiv m\ (d)\}$. Since $\sum_{d|Q} \mu(Q/d) = 0$ we may replace cnt_d by $\psi_d(m) = \mathrm{cnt}_d(m) - q^{N-\deg d}$. For $\deg d \leq N$ every residue class modulo d contains exactly $q^{N-\deg d}$ polynomials of degree $< N$ (the class of m is $\{\tilde m + dg :\ \deg g < N - \deg d\}$), of which one is 0 iff $d \mid m$; hence $\psi_d(m) = -\mathbf{1}_{d|m}$, including $d = 1$, where $\psi_1 = -1$ because $h = 0$ is excluded. For $\deg d > N$ a class contains at most one nonzero h of degree $< N$, namely $\tilde m$ when $0 \leq \deg \tilde m < N$, so $\psi_d(m) = \mathbf{1}_{0\leq \deg \tilde m<N} - q^{N-\deg d}$. Regrouping the moduli $Q = dm$, $(m,d) = 1$, by d (absolute convergence as in Theorem 2) gives $\sum_{Q\neq 1} b(Q) \sum_h c_Q^f(h) = \sum_d W_f(d) \sum_{s,s'} \psi_d(s'-s) + 1$, the $+1$ compensating the term $Q = 1$, i.e. $(d,m) = (1,1)$, of the regrouped sum, whose value is $\psi_1 = -1$. For $\deg d \leq N$, $\sum_{s,s'} \psi_d(s'-s) = -\#\{(s,s') :\ s \equiv s'\ (d)\} = -\omega_f(d)$, which gives $-P(1) - \sum_{1\leq \deg d\leq N} a_f(d)$. For $\deg d > N$ the pairs with $s = s'$ have $\tilde m = 0$ and contribute $-\omega_f(d) q^{N-\deg d}$, the pairs with $s \neq s'$ contribute $\mathrm{Off}_f(N)$. □

Over $\mathbb{Z}$ the modulus $d = 1$ drops out of (8) because $\#\{h \leq H\} = H$ exactly; over $\mathbb{A}$ it leaves the constant $1 - P(1)$ because $h = 0$ is excluded from the q^N polynomials of degree $< N$. The three diagonal terms are the exact analogue of $-C^2 \sum_d a_f(d)\{H/d\}$ in (9): the fractional part $\{H/d\}$, of mean $\frac{1}{2}$ for $d \leq H$ and equal to H/d for $d > H$, is replaced by 1 for $\deg d \leq N$ and by $q^N/|d|$ for $\deg d > N$.

The counterpart of Theorem 3 is the following. Put $T = q^{-s}$. The Dirichlet series $D_f(s) = \sum_d a_f(d)|d|^{-s}$ is a power series in T whose coefficient of T^j is $\sum_{\deg d=j} a_f(d)$, and the local computation of Theorems 3 and 4 goes through verbatim: $D_f(s) = \zeta_{\mathbb{A}}(s+1) L(s+1, \chi_D) E_f(s)$ with $\zeta_{\mathbb{A}}(s+1) = 1/(1-T)$, $L(s+1, \chi_D)$ a polynomial in T, and $E_f(s) = \tilde H_f(s)(1-q^{-1}T^2)^2/L(2s+2, \chi_D)$, where $\tilde H_f$ converges absolutely for $|T| < q$ and the polynomial $L(2s+2, \chi_D)$ has its zeros on $|T| = q^{3/4}$ by Weil's theorem. Hence D_f is meromorphic in $|T| < q^{3/4}$ with a single simple pole at $T = 1$, and $(1-T)D_f \to 1/C(f)$ as $T \to 1$ by the same computation as over $\mathbb{Q}$; therefore $\sum_{\deg d=j} a_f(d) = 1/C(f) + O(q^{-(3/4-\varepsilon)j})$, $\sum_{1\leq \deg d\leq N} a_f(d) = N/C(f) + c_f + O(q^{-(3/4-\varepsilon)N})$ and $q^N \sum_{\deg d>N} a_f(d)/|d| = 1/(C(f)(q-1)) + O(q^{-(3/4-\varepsilon)N})$. Hence over $\mathbb{F}_q[u]$ the analogue of Conjecture 1,

$$\sum_{\substack{h\neq 0\\ \deg h<N}} \big(S_f(h) - C(f)^2\big) = -C(f)\,N + O_f(1), \tag{11}$$

is *equivalent* to $\mathrm{Off}_f(N) = O(1)$: the whole content of Hypothesis (E) sits in the off-diagonal pairs modulo the moduli of degree $> N$. (The coefficient is $-C$ rather than $-\frac{1}{2}C$ because the mean $\frac{1}{2}$ of $\{H/d\}$ is replaced by the exact value 1.) The remainder is more concrete than it looks: its first part is a finite sum.

Lemma 9. *Let $f = t^2 - D$ with $D \in \mathbb{A}$ not a square, and for squarefree d let $A_N(d)$ be the number of ordered pairs (s, s') of roots of f modulo d with $s \neq s'$ whose difference $\tilde{m}$ is nonzero of degree $< N$. Then*

$$\mathrm{Off}_f(N) = \sum_{N<\deg d\le M(N)} W_f(d)\, A_N(d) \;-\; q^N \sum_{\deg d>N} \frac{W_f(d)\big(\omega_f(d)^2 - \omega_f(d)\big)}{|d|},$$

with $M(N) = N - 1 + \max(2N-2, \deg D)$. The series converges absolutely, and it is computable from the exact identities

$$\sum_d \frac{W_f(d)\,\omega_f(d)^2}{|d|} = 1, \qquad \sum_d \frac{W_f(d)\,\omega_f(d)}{|d|} = \prod_P \Big(1 + \frac{\omega_f(P)(1-\omega_f(P))}{(|P| - \omega_f(P))^2}\Big) =: A_f(1),$$

both sums over all squarefree monic d including $d = 1$.

Proof. By definition $\mathrm{Off}_f(N) = \sum_{\deg d>N} W_f(d)\big(A_N(d) - (\omega_f(d)^2 - \omega_f(d))q^{N-\deg d}\big)$, so it remains to bound the degrees with $A_N(d) \neq 0$. Let (s, s'), $s \neq s'$, have $0 \le \deg\tilde{m} < N$, and split $d = d_1 d_2$ with $d_2 = \prod\{P \mid d : \; s \equiv s' \ (P)\}$. Then $d_2 \mid \tilde{m}$, so $\deg d_2 \le \deg\tilde{m} \le N-1$ because $\tilde{m} \neq 0$. For $P \mid d_1$ the two roots are distinct modulo P, hence $P \nmid D$ and $s' - s \equiv \pm 2\sqrt{D}$, so $\tilde{m}^2 \equiv 4D \pmod{d_1}$ and $d_1 \mid \tilde{m}^2 - 4D$; since D is not a square, $\tilde{m}^2 - 4D \neq 0$ and $\deg d_1 \le \deg(\tilde{m}^2 - 4D) \le \max(2N-2, \deg D)$. Adding the two bounds gives $\deg d \le M(N)$. For the identities, $W_f(d)/|d| = b(d)\sum_{(m,d)=1} \mu(m) b(m)\omega_f(m)^2$, so $\sum_d W_f(d)\omega_f(d)^2/|d| = \sum_Q b(Q)\omega_f(Q)^2 \sum_{d\mid Q} \mu(Q/d) = b(1)\omega_f(1)^2 = 1$, and $\sum_d W_f(d)\omega_f(d)/|d| = \sum_Q b(Q) \sum_{d\mid Q} \mu(Q/d)\omega_f(d)\omega_f(Q/d)^2$ is multiplicative in Q with local factor $1 + b_P(\omega_f(P) - \omega_f(P)^2)$; all rearrangements are justified by absolute convergence, since $\sum_Q b(Q)\omega_f(Q)^2 < \infty$. $\square$

So (11) is equivalent to the statement that, over the moduli d of degree between N and $M(N) \approx 3N$, the differences of distinct roots of f modulo d fall among the residues of degree $< N$ with the frequency $q^{N-\deg d}$, up to a bounded total discrepancy; nothing about moduli of degree $\le N$ or $> M(N)$ is involved. Neither term of Lemma 9 is bounded a priori; only their difference is claimed to be. For a prime modulus P of degree k the count asks whether the residue $2\sqrt{D} \bmod P$ has degree $< N$; written through the additive characters trivial on $\mathbb{A}_{<N}$, this is a Weyl sum of the roots of a quadratic congruence over primes of fixed degree, the function-field analogue of the sums of Duke, Friedlander and Iwaniec, and for fixed N and $q \to \infty$ each of the finitely many degrees in Lemma 9 is an algebraic count with square-root cancellation by Deligne's theorem [14], as in the work of Bary-Soroker and Entin on prime values of polynomials over $\mathbb{F}_q[u]$ [3, 6]. We have not written this out; the large-q form of (11), with $O(q^{-1/2})$ in place of $O(1)$, is where we expect the first theorem beyond the linear case, and Lemma 9 says exactly which finite family of counts it requires.

Exact computation. Theorem 8 and Lemma 9 can be checked exactly, and we did so in two independent ways. First (`thesis/ff.ts`), for each nonzero h of degree $< N$, $S_f(h)/(C^2P(1))$ is a finite product over the primes dividing $h(h^2 - 4D)$; the two diagonal sums are computed from the local data (the series through $A_f(1)$), and $\mathrm{Off}_f(N)$ follows from the identity by subtraction. Second (`thesis/ff-tail.ts`), $\mathrm{Off}_f(N)$ is computed from Lemma 9: the finite sum from the roots of f modulo every squarefree d of degree $\le M(N)$ with all prime factors split or ramified, assembled by the Chinese remainder theorem, and the series from the two identities; for the 15 smallest cases the finite sum was recomputed by brute force (all monic d, roots by exhaustive search), with agreement to $4 \cdot 10^{-16}$. Tables 4 and 5 report $f = t^2 - D$ for $q = 3, 5, 7$ and 6 choices of D. The two computations of $\mathrm{Off}_f(N)$ agree to $3 \cdot 10^{-5}$ (the residual is the truncation of the Euler products at the prime table); the increments of $\sum_{\deg d\le N} a_f(d)$ converge to $1/C(f)$ and the diagonal tail to $1/(C(f)(q-1))$ as predicted (after a transient at $N \le 2$ for $q = 3$, where a prime of norm 3 with $\omega_f = 2$ makes $P_P = -3$ and $P(1) < 0$); and $\mathrm{Off}_f(N)$ is bounded, $|\mathrm{Off}_f(N)| \le 1.83$ in all cases with a variation over N of at most 1.13, with no sign of growth.

Table 4: The function-field identity of Theorem 8 for $f = t^2 - D$ over $\mathbb{F}_q[u]$ (`thesis/ff.ts`): the left side (LHS), $\sum_{1\le \deg d\le N} a_f(d)$ and its increment from $N-1$ to N (limit $1/C(f)$), the diagonal tail $q^N \sum_{\deg d>N} a_f(d)/|d|$ (limit $1/(C(f)(q-1))$), and the off-diagonal remainder $\mathrm{Off}_f(N)$, conjectured $O(1)$.

q	D	$C(f)$	$1-P(1)$	N	$\#h$	LHS	$\sum a_f$	incr.	diag. tail	$\mathrm{Off}_f(N)$
3	u	0.851	2.999	1	2	-2.000	1.999	–	1.168	-1.832
				2	8	-3.202	5.198	3.199	0.305	-0.699
				3	26	-4.758	5.302	0.104	0.810	-1.645
				4	80	-5.855	7.181	1.879	0.551	-1.122
				5	242	-7.167	8.308	1.126	0.527	-1.331
				6	728	-8.201	9.252	0.945	0.637	-1.311
				7	2186	-9.548	10.587	1.334	0.576	-1.385
3	u^2+1	1.459	3.857	1	2	-2.000	5.714	–	-0.094	-0.237
				2	8	-2.286	5.306	-0.408	0.127	-0.710
				3	26	-2.921	5.129	-0.177	0.558	-1.092
				4	80	-3.530	6.522	1.394	0.279	-0.586
				5	242	-4.214	6.994	0.472	0.366	-0.711
				6	728	-4.925	7.770	0.776	0.322	-0.691
				7	2186	-5.626	8.384	0.615	0.351	-0.748
3	u^2+2	1.703	0.500	1	2	-0.801	1.000	–	0.332	0.030
				2	8	-1.803	1.699	0.700	0.295	-0.309
				3	26	-2.019	2.273	0.574	0.311	0.065
				4	80	-2.827	2.924	0.651	0.282	-0.121
				5	242	-3.372	3.476	0.551	0.294	-0.103
				6	728	-3.949	4.061	0.585	0.297	-0.091
				7	2186	-4.528	4.661	0.600	0.292	-0.075
5	u	0.956	0.721	1	4	-1.768	1.209	–	0.366	-0.914
				2	24	-2.958	2.803	1.594	0.237	-0.638
				3	124	-3.770	3.728	0.925	0.262	-0.501
				4	624	-4.733	4.774	1.046	0.266	-0.414
				5	3124	-5.769	5.845	1.071	0.258	-0.388
5	u^2+2	1.247	0.701	1	4	-0.978	1.197	–	0.287	-0.194
				2	24	-2.250	2.493	1.296	0.142	-0.316
				3	124	-2.786	2.986	0.493	0.215	-0.286
				4	624	-3.701	3.857	0.871	0.202	-0.342
				5	3124	-4.465	4.671	0.813	0.197	-0.298
7	u	0.979	0.434	1	6	-1.474	1.245	–	0.196	-0.468
				2	48	-2.392	2.451	1.207	0.167	-0.208
				3	342	-3.478	3.455	1.003	0.169	-0.288
				4	2400	-4.439	4.468	1.013	0.171	-0.235

Table 5: Lemma 9 checked (`thesis/ff-tail.ts`): the finite sum $\sum_{N<\deg d\leq M(N)} W_f(d)A_N(d)$ over the $\#d$ squarefree moduli with all prime factors split or ramified, the series $q^N \sum_{\deg d>N} W_f(d)(\omega_f(d)^2 - \omega_f(d))/|d|$, their difference, and $\mathrm{Off}_f(N)$ obtained independently from the $S_f(h)$ through Theorem 8 (Table 4).

q	D	N	$M(N)$	$\#d$	finite sum	series	difference	$\mathrm{Off}_f(N)$ from LHS
3	u	1	1	0	0.0000	1.8322	-1.8322	-1.8322
		2	3	6	1.5993	2.2980	-0.6988	-0.6988
		3	6	199	1.9465	3.5913	-1.6448	-1.6448
		4	9	4726	1.7928	2.9146	-1.1218	-1.1218
		5	12	112209	2.3993	3.7307	-1.3314	-1.3314
3	u^2+1	1	2	1	0.0000	0.2367	-0.2367	-0.2367
		2	3	5	0.0000	0.7102	-0.7102	-0.7102
		3	6	146	1.2164	2.3082	-1.0918	-1.0918
		4	9	3500	0.9700	1.5557	-0.5857	-0.5857
		5	12	83301	1.1099	1.8208	-0.7110	-0.7110
3	u^2+2	1	2	2	0.1999	0.1696	0.0303	0.0303
		2	3	6	0.0000	0.3088	-0.3088	-0.3088
		3	6	199	0.4179	0.3528	0.0650	0.0650
		4	9	4726	0.2862	0.4071	-0.1209	-0.1209
		5	12	112209	0.4283	0.5308	-0.1025	-0.1025
5	u	1	1	0	0.0000	0.9141	-0.9141	-0.9141
		2	3	33	0.1063	0.7447	-0.6384	-0.6384
		3	6	4032	0.7041	1.2048	-0.5006	-0.5007
5	u^2+2	1	2	5	0.6271	0.8215	-0.1943	-0.1943
		2	3	28	0.1140	0.4301	-0.3160	-0.3160
		3	6	3314	0.6868	0.9730	-0.2862	-0.2862
7	u	1	1	0	0.0000	0.4679	-0.4679	-0.4679
		2	3	96	0.3520	0.5597	-0.2077	-0.2077
		3	6	29328	0.4140	0.7022	-0.2882	-0.2882

4 Where the zeros of $L(s,\chi_D)$ enter, and why they are hard to see

This section is about lower-order terms. It explains the mechanism by which the zeros of ζ and of $L(s,\chi_D)$ enter the fluctuations of $\sum_{h\le H} S_f(h)$, exactly as the zeros of ζ enter the twin-prime case, and reports an honest negative result: at the ranges we can compute they are not yet visible.

4.1 The explicit formula for the diagonal

For $f(t)=t$ the mechanism is known: Vaughan [23] proved $\sum_{k\le x}(x-k)\mathfrak{S}(k)=\frac{1}{2}x^2-\frac{1}{2}x\log x+\frac{1}{2}(1-\gamma-\log 2\pi)x+E(x)$ with $E(x)=\Omega_\pm(x^{1/4})$, and Goldston and Suriajaya [11] showed that for the Riesz means $\sum_{k\le x}(x-k)^m\mathfrak{S}(k)$, $m\ge 2$, the error term is $x^{m-1}\sum_{|\gamma|\le U} a(\rho)x^{\rho/2}+O(x^{m-1+\varepsilon})$ over the zeros $\rho=\beta+i\gamma$ of ζ, the zeros entering through the factor $1/\zeta(2s)$ of the generating Dirichlet series, and that unconditionally $E_m(x)=\Omega_\pm(x^{m-3/4})$. Theorem 4 puts the diagonal part of our sum in exactly this situation. Let $A_f=a_f*1$, i.e. $A_f(n)=\sum_{d\mid n}a_f(d)$, so that the diagonal part of $\sum_{h\le H}(S_f(h)-C^2)$ is $-C^2\sum_d a_f(d)\{H/d\}=C^2\big(\sum_{n\le H}A_f(n)-H\sum_d a_f(d)/d\big)$ and $\sum_n A_f(n)n^{-s}=\zeta(s)D_f(s)$.

Proposition 10. *Let f be an irreducible quadratic with discriminant D, $m\ge 2$, and assume the Riemann hypothesis for ζ and for $L(s,\chi_D)$. Then there is $\varepsilon_0>0$ such that*

$$\sum_{n\le x}(x-n)^m A_f(n)=P_m(x)+c_f\,x^{m-2/3}+x^{m-1}\sum_{\substack{\rho\\|\gamma|\le U}} a_m(\rho)\,x^{\rho/2}+O(x^{m-3/4-\varepsilon_0}),$$

for some $x^5\le U\le 2x^5$, where P_m collects the residues at $s=1$ (x^{m+1}) and $s=0$ ($x^m\log x$ and x^m), $c_f\,x^{m-2/3}$ is the residue at the pole $s=-\frac{2}{3}$ of $\zeta(3s+3)$, ρ runs over the zeros of $\zeta(s)^2L(s,\chi_D)$, and $a_m(\rho)$ is the residue of

$$\frac{m!\,\zeta(s)\zeta(s+1)L(s+1,\chi_D)\zeta(3s+3)L(3s+3,\chi_D)\tilde{H}_f(s)}{\zeta(2s+2)^2L(2s+2,\chi_D)\,s(s+1)\cdots(s+m)}$$

at $s=\rho/2-1$ (a polynomial of degree one in $\log x$ at the double zeros of ζ).

Sketch. By Theorem 4 the generating series $\zeta(s)D_f(s)$ of A_f is meromorphic in $\Re s>-1$; in the strip $-\frac{3}{4}-\varepsilon_0\le\Re s\le 2$, for $\varepsilon_0<\frac{1}{20}$, its poles are exactly $s=1$, $s=0$, $s=-\frac{2}{3}$ and the points $s=\rho/2-1$ (the next real pole is at $-\frac{4}{5}$ and the next family of zero-induced poles, from $\zeta(4s+4)L(4s+4,\chi_D)^2$, lies on $\Re s=-\frac{7}{8}$). The proof of [11, Theorem 1] applies with $\zeta(2s+2)^2L(2s+2,\chi_D)$ in place of $\zeta(2s)$: the contour is moved to $\Re s=-\frac{3}{4}-\varepsilon_0$ at a height U chosen, by a zero-density estimate for ζ and $L(s,\chi_D)$, so that $1/(\zeta^2L)$ is not large on the horizontal segments, and the Riesz weight of order $m\ge 2$ makes the vertical integral absolutely convergent; on the vertical line, $\Re(2s+2)=\frac{1}{2}-2\varepsilon_0$ lies just left of the critical line, and the Riemann hypothesis with the functional equation gives $1/\zeta(2s+2)$, $1/L(2s+2,\chi_D)\ll|t|^{2\varepsilon_0+\varepsilon}$ there, while $\tilde{H}_f$ is bounded. Unlike the linear case, the contour cannot be moved across the whole critical strip of $\zeta(2s+2)$, because $\Re s=-1$ is a natural boundary; this is why the statement is conditional and why the error term is $x^{m-3/4-\varepsilon_0}$ rather than $x^{m-1+\varepsilon}$. (Moving the contour further, to $\Re s=-\frac{7}{8}-\varepsilon_0$, is possible but useless: $|D_f|$ grows towards the natural boundary and the coefficients of the second family of zero terms are individually enormous, cancelling against the line integral.) We state this as a proposition rather than a theorem because we have transplanted the argument and checked the new ingredients (the location of the poles, the growth of $\tilde{H}_f$, the exponents), but have not written out the contour estimates line by line. $\square$

Since $A_f(n)\ge 0$ for $f=t^2+1$, Landau's theorem applies as in [11, Theorem 4] and gives, unconditionally, $\Omega_\pm(x^{m-3/4})$ for the error term after $P_m(x)+c_fx^{m-2/3}$ (the pole at $-\frac{2}{3}$ is real and must be subtracted first; the first non-real singularity is at $-\frac{3}{4}+3.01i$, from the zero $\frac{1}{2}+6.02i$ of $L(s,\chi_{-4})$). Thus the fluctuations of the diagonal part of the pair-correlation sum of t^2+1 are governed by the zeros of ζ and of $L(s,\chi_{-4})$, the Dirichlet L-function of $\mathbb{Q}(i)$: this is the precise sense in which the field cut out by a polynomial enters the second moment of its prime values. Under the Riemann Hypothesis for ζ and $L(s,\chi_{-4})$ the diagonal error after the smooth terms is $O(x^{m-3/4+\varepsilon})$; a zero of real part $\beta > \frac{1}{2}$ would produce a term of size $x^{m-1+\beta/2}$.

4.2 A numerical search

The proposition invites a test: compute the Riesz means, remove the main terms, and look in the residual for oscillations $\cos(\frac{\gamma}{2}\log x)$. We did this for $f = t^2+1$ in two versions (`thesis/riesz.ts`, `thesis/riesz-diag.ts`, `thesis/riesz-fit.ts`).

Full sum. By Theorem 5, $S_f(h)$ is computable exactly by sieving h and h^2+4; we did so for all $h \leq 40{,}000{,}000$, formed $R_m(x) = \sum_{h\leq x}(x-h)^m S_f(h)$ at 4096 points equally spaced in $u = \log x \in [\log 10^4, \log 40{,}000{,}000]$, removed the smooth terms by least squares and analysed the residual, normalised by $x^{m-3/4}$, by a windowed Fourier transform in u (frequency resolution 0.76). The zeros of $L(s,\chi_{-4})$ were located independently ($\gamma = 6.0209, 10.2438, 12.9881, \ldots$). The result is a clear null: after detrending the spectrum, the power at the predicted frequencies $\gamma/2$ is statistically indistinguishable from the power at control and at random frequencies. This is exactly what Theorem 6 predicts. The off-diagonal part contributes $O(H)$ to the Cesàro sum, and it fluctuates: it is a sum of fractional parts over the differences of the roots of f modulo d, and these fluctuations are of size x in R_1, against $x^{1/4}$ for the zero terms of the diagonal. The zeros are buried under the off-diagonal.

Diagonal sum. Proposition 10 concerns the diagonal alone, which is computable exactly (via $\omega_f(p) = 1+\chi_{-4}(p)$) for $n \leq 20{,}000{,}000$. Here the residual after the smooth terms has the size the theorem predicts ($\approx 0.1\,x^{m-3/4}$ in amplitude), and its strongest spectral peaks lie near predicted frequencies (Table 6), but the predicted frequencies are dense (about one per 0.8 in ω) while the resolution is 0.83, so proximity proves little. Two sharper tests are both negative at this range: the detrended power at the predicted frequencies in the band $4 \leq \omega \leq 24$ averages 0.51 (zeros of L) and 0.63 (zeros of ζ) in units of the spectrum's scatter, against 0.57 ± 0.76 at random frequencies in the same band; and a matched filter, fitting cosines and sines at $\gamma/2$ for the first 6 zeros of $L(s,\chi_{-4})$ and 4 of ζ, explains 13.5% of the residual variance, while the same set of frequencies shifted by 0.15–0.6 explains 14.1% on average and up to 17.9%.

Table 6: Strongest peaks of the spectrum of the diagonal residual ($m = 2$, $x \leq 20{,}000{,}000$) and the nearest predicted frequency $\gamma/2$. Resolution 0.83; the predicted frequencies have mean spacing ≈ 0.8.

#	ω	nearest	$\gamma/2$	$\lvert\Delta\rvert$
1	7.22	ζ	7.07	0.15
2	5.16	$L(s,\chi_{-4})$	5.12	0.04
3	10.70	$L(s,\chi_{-4})$	10.73	0.03
4	12.74	$L(s,\chi_{-4})$	12.86	0.12
5	24.08	ζ	24.00	0.08
6	20.28	$L(s,\chi_{-4})$	20.16	0.12
7	8.88	$L(s,\chi_{-4})$	9.15	0.27
8	15.02	$L(s,\chi_{-4})$	14.83	0.19

So the zeros are not detected at $x \leq 2\cdot 10^7$ by this method, and the reason is instructive. The fitted smooth basis did not contain the term $c_f x^{m-2/3}$ of Proposition 10, which is larger than the zero terms $|a_m(\rho)|x^{m-3/4}$, and the $\gamma/2$-spacing set by the length of the $\log x$ range is comparable to the spacing of the predicted frequencies. In the sequel we compute the main terms, c_f and the coefficients $a_m(\rho)$ exactly and subtract them; the residual then agrees with the predicted sum over the zeros of ζ and $L(s,\chi_{-4})$ in amplitude and phase (regression coefficient 1.01 for $m=2$, with the two families separately at 1.006 and 1.020). Without the exact coefficients one would need a range $x \geq 10^{10}$, beyond a direct sieve of $A_f(n)$. We record the negative result of the fitting method here; the exact computation is the first step of the sequel.

4.3 The Galois group

The off-diagonal Dirichlet series $\sum_d d^{-s} W_f(d) \sum_{s\neq s'} \zeta(s, \langle (s'-s)/d\rangle)$ (Hurwitz zeta functions of the root differences) has, at $s=0$, a polar part of order at most $r_f - 2$ times the bias of the distribution of the root differences, where r_f is the number of orbits of the Galois group on ordered pairs of roots; $r_f = 2$ exactly when the group is 2-transitive. For $r_f \geq 3$ (cyclic cubics, D_4, C_4 and V_4 quartics) a biased distribution of the differences of distinct roots modulo d could therefore add a logarithm to $\Sigma_f(H)$ and break the universality of $-\frac{1}{2}$. We tested this on 20 polynomials with non-2-transitive groups (Shanks' simplest cubics $t^3 - at^2 - (a+3)t - 1$, $t^4 - c$, and C_4, V_4 quartics; the group was verified from the distribution of $\omega_f(p)$) against 9 with S_3 or S_4 (Table 7; `thesis/galois.ts`). The slope of $\Sigma_f(H)/C$ over $10^3 \leq H \leq 10^5$ averages -0.43 ± 0.19 for the non-2-transitive groups and -0.45 ± 0.05 for the 2-transitive ones. The means agree with each other and with the finite-H value expected from Section 3.5; the scatter between individual polynomials is 3–5 times larger when the group is not 2-transitive, which is what a higher-order polar part with fluctuating coefficient produces. The constant $-\frac{1}{2}$ appears to be independent of the Galois group; the rate of approach to it is not.

Table 7: Slope of $\Sigma_f(H)$ per unit of $\log H$ over $10^3 \leq H \leq 10^5$, divided by $C(f)$, by Galois group (mean $\pm$ s.d. over the polynomials of the group). r_f is the number of orbits on ordered pairs of roots. Conjecture 1 predicts -0.50 asymptotically.

group	r_f	n	mean C	slope/C	s.e.
C_3	3	8	2.86	-0.380 ± 0.148	0.052
S_3	2	5	1.54	-0.459 ± 0.066	0.030
D_4	3	6	1.21	-0.532 ± 0.178	0.073
C_4/V_4	4	6	2.10	-0.394 ± 0.204	0.083
S_4/A_4	2	4	1.69	-0.442 ± 0.030	0.015

5 Primes along polynomial sequences

Summary: primes along polynomial sequences fluctuate less than a random model of the same density, and the deficit matches the pair-correlation prediction cell by cell (Table 8). Everything in this section is conditional on the Hardy–Littlewood pair conjecture for f, in the sense explained in the introduction: we test whether primes behave as (5) predicts when Conjecture 1 is inserted. With $\mathbf{1}_t$, p_t, w_t, A_T, V_T as in the introduction and $\mu_T = \sum_{t<T} p_t$, the prediction is

$$\operatorname{Var}(A_T) \approx V_T + 2\sum_{h=1}^{T-1} \big(S_f(h) - C(f)^2\big) W_T(h), \qquad W_T(h) = \sum_{t<T-h} w_t w_{t+h}, \tag{12}$$

which is (5); for $f(t) = t$, inserting $\sum_{h\leq H} \mathfrak{S}(h) = H - \frac{1}{2}\log H + O(1)$ with $w_t \approx 1/\log X$ gives (4).

Design. For a polynomial f and a length T we count the primes A among $f(t_0), \ldots, f(t_0+T-1)$ and form the dispersion $\Phi = (A-\mu)^2/V$ with $\mu = \mu_T$, $V = V_T$ as above; averaged over the polynomials of a cell, $\Phi = 1$ for independent trials with the Bateman–Horn probabilities, $\Phi < 1$

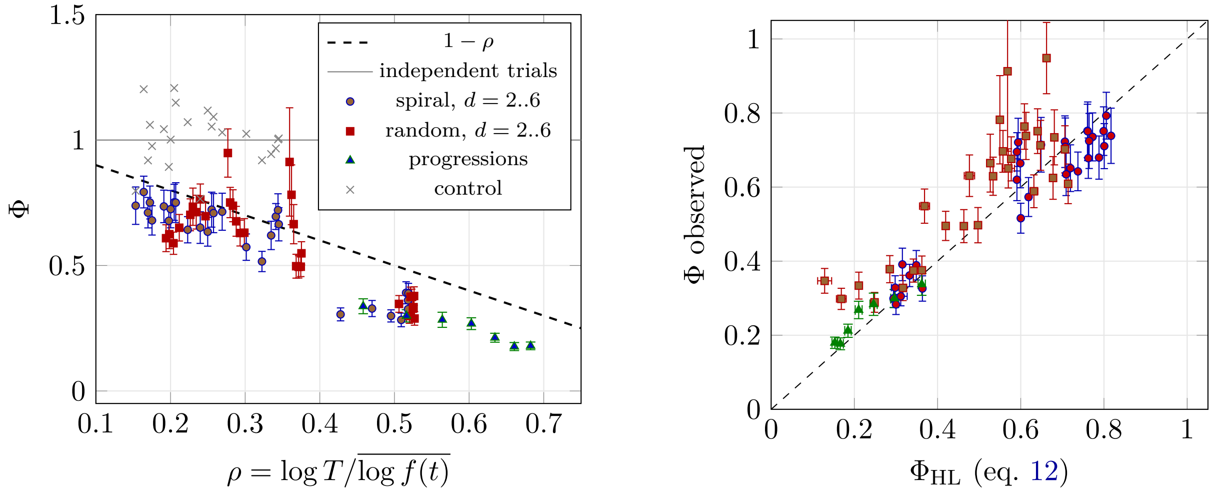

Figure 3: Left: observed dispersion of prime counts along polynomial sequences against $\rho = \log T/\overline{\log f}$ (within a degree $\rho \to 1/d$, so degrees form clusters); the control stays at 1, the sequences lie below 1 and, for degree 2 and the random families, well below $1-\rho$. Right: the same cells against the pair-correlation prediction.

for a variance deficit. Φ_{HL} is the average of $\mathrm{Var}_{\mathrm{HL}}/V$ from (12). Three families are used, 240 polynomials each per degree: arithmetic progressions ($2 \le q \le 300$); spiral polynomials of degree 2 to 6 with base point in a box of radius ≤ 20; random integer polynomials of degree 2 to 6 as above. T runs over $25, 50, \ldots$ up to 3200 ($d \le 2$), 1600 ($d = 3$), 800, 400, 200; the values reach 10^{16} and primality is decided by Miller–Rabin with the first twelve prime bases (deterministic below $3.3 \cdot 10^{23}$). The Euler products for C and S_f are truncated at $p \le 20000$ ($d \le 2$), 8000 ($d = 3$), 3000 otherwise. Standard errors are bootstrap over the polynomials of a cell. As a control, for every polynomial and T the same number of integers is drawn uniformly at random from $[f(t_0), f(t_0+T)]$ and passed through the identical primality and weighting code; its dispersion Φ_{rand} must be 1. Everything is generated by `thesis/variance.ts` from a fixed seed.

Table 8: Variance experiment, averaged over the T-grid of each family and degree (weighted by $1/\mathrm{se}^2$). $\rho = \log T/\overline{\log f}$, so $1-\rho$ is the law (4) with $H = T$; Φ_{HL} is the pair-correlation prediction (12); κ and κ_C are the means of $\Sigma_f(T)/(C^2 \log T)$ and $\Sigma_f(T)/(C \log T)$ over the polynomials of the cell (Conjecture 1: $\kappa_C \to -\frac{1}{2}$); obs/pred is the ratio of the total number of primes to the Bateman–Horn total. All cells are listed in Appendix A.

family	cells	ρ	$1-\rho$	Φ	Φ_{HL}	Φ_{rand}	κ	κ_C	obs/pred
arith. progressions	7	0.588	0.41	0.22 ± 0.01	0.22 ± 0.00	0.98	-0.47	-0.66	0.992
spiral, $d = 2$	8	0.497	0.50	0.32 ± 0.01	0.33 ± 0.00	1.04	-0.59	-0.65	1.011
random, $d = 2$	8	0.522	0.48	0.33 ± 0.01	0.30 ± 0.00	0.98	-0.62	-0.64	0.991
spiral, $d = 3$	7	0.323	0.68	0.62 ± 0.02	0.60 ± 0.00	0.98	-0.67	-0.66	1.004
random, $d = 3$	7	0.368	0.63	0.53 ± 0.02	0.51 ± 0.00	1.00	-0.66	-0.64	0.976
spiral, $d = 4$	6	0.237	0.76	0.67 ± 0.02	0.72 ± 0.00	1.00	-0.81	-0.66	0.988
random, $d = 4$	6	0.287	0.71	0.70 ± 0.02	0.61 ± 0.00	1.02	-0.68	-0.63	0.996
spiral, $d = 5$	5	0.196	0.80	0.72 ± 0.03	0.77 ± 0.00	1.08	-0.64	-0.65	1.008
random, $d = 5$	5	0.236	0.76	0.72 ± 0.03	0.65 ± 0.00	1.04	-0.68	-0.65	0.996
spiral, $d = 6$	4	0.165	0.84	0.75 ± 0.03	0.81 ± 0.00	0.99	-0.69	-0.66	1.008
random, $d = 6$	4	0.202	0.80	0.62 ± 0.03	0.68 ± 0.00	1.00	-0.70	-0.65	0.957

Results. Every one of the 60 polynomial cells has $\Phi < 1$ ($\chi^2 = 9002$ against $\Phi = 1$), while the 67 control cells average 1.009 (range 0.77–1.24) and the total prime counts agree with Bateman–

Horn to 0.904–1.028 (Table 8, Figure 3). The deficit is therefore not an artefact of the primality test, of the weights, or of a biased $C(f)$ (a bias in C inflates Φ and can only hide a deficit). The pair-correlation prediction (12) reproduces the observed dispersion cell by cell to within a few hundredths (Figure 3, right): $\chi^2 = 189.6$ on 60 polynomial cells and 12.3 on the 7 progression cells with statistical errors only, 11 cells being off by more than 2σ. The residuals are of the size of the systematic uncertainty of Φ_{HL} from the truncation of S_f (below), and the largest ones occur for the random quadratics at $T \leq 50$, whose values are in the tens and hundreds, where the asymptotic formula is not expected to apply. The law $1-\rho$ with $H = T$ does not fit ($\chi^2 = 647.7$); the observed deficits are larger, most strikingly for quadratics ($\Phi = 0.28$–0.39 at $\rho \approx 0.51$). This is what Conjecture 1 predicts at finite T: the cell values of $\kappa_C = \Sigma_f(T)/(C \log T)$ lie between -0.66 and -0.63 (progressions: -0.66), i.e. $\Sigma_f(T)$ is below $-\frac{1}{2}C \log T$ by about $1.1\,C$ at $T \approx 10^3$, consistent with Figure 1, and the cell values move slowly towards $-\frac{1}{2}$ as T grows (Appendix A); Table 1 shows that $\Sigma_f(H) + \frac{1}{2}C \log H$ has drifted to $\approx -0.34\,C$ only by $H = 10^5$. The excess deficit is the lower-order term of (6), and it is polynomial-dependent: the two families of the same degree sit at visibly different Φ for the same ρ in Figure 3.

A computational warning. In a first run the pair series of the quadratics was truncated at $p \leq 5000$; the predicted variances at $T \geq 1600$ then came out negative for many polynomials. With $p \leq 20000$ they are positive and correct. $\Sigma_f(H)$ inherits the conditional convergence of the Euler product, and for five test quadratics moving the truncation from 5000 to 60000 still changes $\Sigma_f(3200)$ by 5–20% (`thesis/trunc.ts`); numerical work on second moments of prime values of polynomials must control this (see [24] for tail estimates).

6 Polynomials of the d-dimensional Ulam spiral

This section records the origin of the project; the reader interested only in the analytic results can skip it. Our "spiral" families come from the following construction, which is of independent interest because it turns the familiar Ulam picture [20] into a generator of polynomial families in every degree.

Definition 11. *Let $\sigma_2 : \mathbb{N} \to \mathbb{Z}^2$ be the Ulam spiral ($\sigma_2(1) = (0,0)$, $\sigma_2(2) = (1,0)$, $\sigma_2(3) = (1,1)$, counter-clockwise), so that ring k carries the integers $(2k-1)^2 + 1, \ldots, (2k+1)^2$. For $d \geq 3$, shell k of $\mathbb{Z}^d$ (Chebyshev norm k) carries $(2k-1)^d + 1, \ldots, (2k+1)^d$: the $(d-1)$-dimensional shell of radius k is swept through the new axis x_d in the layer order $0, +1, -1, \ldots, \pm(k-1)$, each layer in the order of σ_{d-1}, and the caps $x_d = \pm k$ are then filled with the $(d-1)$-dimensional ball of radius k in the order of σ_{d-1}.*

Theorem 12. *For $b \in \mathbb{Z}^d$ and $v \in \{-1,0,1\}^d \setminus \{0\}$ there are $t_0 \leq 2\max_i |b_i| + 2$ and $f_{b,v} \in \mathbb{Q}[t]$ of degree exactly d, with leading coefficient 2^d and denominator dividing $d!$, such that the integer at $b + tv$ equals $f_{b,v}(t)$ for all $t \geq t_0$.*

Proof. For $t > 2\max|b_i| + 1$ the order of the $|b_i + tv_i|$ is fixed, so the shell index $k(t) = \max_i |b_i + tv_i|$ is linear in t with slope 1, and the point stays in one face (band or cap) of the shell with a fixed layer-sign. By induction on d the offset of the point within its shell is a polynomial in k of degree at most $d-1$: for $d = 2$ it is the position along one ring side; for $d \geq 3$ it is $\ell\, s_{d-1}(k) + \mathrm{off}_{d-1}$ in a band, with ℓ the layer index (linear in t) and $s_{d-1}(k) = (2k+1)^{d-1} - (2k-1)^{d-1}$, and $(2k-1)s_{d-1}(k) + \varepsilon(2k+1)^{d-1} + n_{d-1} - 1$ in a cap, where off_{d-1} and n_{d-1} are the corresponding quantities of the projected ray. Hence the integer is $(2k(t)-1)^d + O(k^{d-1})$, of degree d in t with leading coefficient 2^d; the coefficients are rational, and the denominator of an integer-valued polynomial of degree d divides $d!$. $\square$

Before t_0 a ray runs through a lower-degree transient inside one shell and possibly through other degree-d regimes; only the eventual regime governs the far part of the line. Our detector locates it by exact finite differences at $t \geq 2\max|b_i| + 2$ and was checked on $5,070$ random rays in dimensions 2–7. The spiral was the starting point of this work: the first author asked whether its "pseudo-patterns" become polynomial in higher dimensions, whether the Fibonacci dimensions are special, and whether the patterns might be quasiperiodic. Theorem 12 answers the first question; the other two have negative answers, documented in the repository (`thesis/REPORT-full.md`): along random rays the number of primes agrees with Bateman–Horn within one standard error in every dimension, the line-to-line dispersion of prime counts inside a box is reproduced by the spread of the constants $C(f)$ alone, the Fibonacci dimensions are indistinguishable from their neighbours, and the diffraction pattern of the primes on the spiral consists of rational Bragg peaks over a diffuse background. The visible pattern of a prime spiral in any dimension is the function line $\mapsto C(f_{\text{line}})$ and nothing else; the one thing the spiral revealed that is not in the constants is the variance deficit of Sections 3 and 5, a property of polynomial sequences and not of the geometry.

7 Discussion

Summary. The message of the paper in one sentence: the fluctuation of the number of prime values of a polynomial is governed, at second order, by an explicit arithmetic function whose generating series is the Dedekind zeta function of the field cut out by the polynomial, and the resulting variance deficit is $\frac{1}{2}C(f)\log H$, universal in form and linear in $C(f)$.

What is proved and what is not. What is proved is Theorems 2–6, 8 and 12: the exact identity, the Dedekind-zeta generating function of the diagonal with residue $1/C(f)$, the symmetric-square L-function inside it, the closed formula for quadratics, the Cesàro form of Conjecture 1 for linear f and its reduction for general f to $\mathrm{Off}^*_f = o(H\log H)$ (Hypothesis (E) for the constant term), and the vanishing of the small-modulus off-diagonal over $\mathbb{F}_q[u]$ together with the finite support of what remains (Lemma 9). Proposition 10 rests on transplanting the proof of Goldston and Suriajaya, which we have sketched but not written out. What is numerical is the rest: for 25 quadratics computed exactly to $H = 10^6$ with C spanning more than a decade the slope of Σ_f scales as C^α with $\alpha = 0.87 \pm 0.09$ (1.10 ± 0.11 unweighted), excluding the C^2 law; the coefficient is about 0.82 of the asymptotic $\frac{1}{2}$ at these H, as the drift of A_f predicts; the off-diagonal remainder of quadratics is bounded to 10^5 and the function-field tail to $N = 7$; and the prime counts along polynomial sequences have the variance that (12) predicts, with the finite-T excess of the deficit accounted for by the constant terms A_f.

Three consequences and open problems. (i) The Montgomery–Soundararajan law $\mathrm{Var}/\mathbb{E} \to 1 - \log T/\log X$ is universal for polynomial prime values, but the approach to it is slow and polynomial-dependent; any numerical or heuristic study of the second moment at accessible T must include A_f. The smoothed diagonal gives A_f in terms of the constant B_f of Theorem 3, i.e. of $E'_f(0)/E_f(0)$ and the residue of ζ_K, up to the $O(1)$ contribution of the off-diagonal. (ii) Proving the leading term of Conjecture 1 for a 2-transitive Galois group is now the problem of saving an unbounded factor over the trivial bound for Off^*_f (Remark 7); the constant term needs Hypothesis (E). For quadratics the roots modulo d are controlled by Kloosterman sums (Hooley; Duke, Friedlander and Iwaniec), and this is the case to attempt first; over $\mathbb{F}_q[u]$ Theorem 8 reduces everything to the moduli of degree $> N$, where Deligne's theorem applies as $q \to \infty$. (iii) The Dedekind zeta function. For $f(t) = t$ the error term in the Riesz means of $\sum_{h\leq x}\mathfrak{S}(h)$ is an explicit formula over the zeros of ζ [11], the zeros entering through the factor $1/\zeta(2s)$ that the squarefree support of the Ramanujan-sum expansion produces. The expansion (7) has the same shape, with $D_f(s) = \zeta_K(s+1)E_f(s)$ generating the diagonal and the squarefree support producing, inside E_f, Euler factors of the form $1 - c_p\,p^{-2-2s}$ with c_p a quadratic expression in $\omega_f(p)$; working out the corresponding explicit formula would show that the fluctuations of the pair-correlation sum, hence of the conjectured variance of the prime values of f, are governed by the zeros of ζ_K (and of the L-function of the permutation representation of the Galois group of f on pairs of roots, which enters through the p^{-2} terms of the local factors). This is the precise sense in which a Riemann hypothesis, that of the field cut out by a lattice line's polynomial, governs the primes on that line. By contrast a direct analogue of the Goldston–Montgomery equivalence [9] is not available, because prime values of a non-linear polynomial are not the support of a Dirichlet series. Finally, the framework of [18] predicts Gaussian fluctuations only for $H \leq X^{1-\delta}$; along a polynomial sequence T is at most $X^{1/d}$, so the restriction is automatically satisfied, and the higher moments should be computable by the same means.

Data and code

All code (TypeScript), the raw results and the scripts that generate every number, table and figure of this paper from them are in the repository accompanying the manuscript: `thesis/sumS.ts`, `thesis/exact.ts`, `thesis/offdiag.ts`, `thesis/ff.ts`, `thesis/ff-tail.ts` (Section 3), the `riesz` scripts and `thesis/galois.ts` (Section 4), `thesis/variance.ts` (Section 5), `thesis/run.ts` (Section 6 and the negative results), `paper/gen-macros.ts` (tables). Random seeds are fixed.

Statement on authorship and the use of AI

The questions about the Ulam spiral that started this work, and the decision to pursue the variance phenomenon, were the first author's. The construction of Definition 11, the proof of Theorem 12, the heuristic of Section 3.1, the software, the experiments and the text were produced by the Anthropic model Claude Fable 5.1 working interactively under the first author's direction in September 2026. The first author has verified the code and the results and takes responsibility for the content.

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A All cells of the variance experiment

family	d	T	ρ	$\bar{\mu}$	Φ	Φ_{HL}	Φ_{rand}	κ	obs/pred
AP	1	50	0.458	9.9	0.34 ± 0.03	0.36 ± 0.01	0.91	-0.52	0.990

family	d	T	ρ	$\bar{\mu}$	Φ	Φ_{HL}	Φ_{rand}	κ	obs/pred
AP	1	100	0.516	18.9	0.30 ± 0.02	0.30 ± 0.01	0.89	-0.53	0.988
AP	1	200	0.564	35.9	0.28 ± 0.03	0.25 ± 0.01	1.04	-0.48	0.986
AP	1	400	0.603	67.9	0.27 ± 0.02	0.21 ± 0.01	1.13	-0.48	0.991
AP	1	800	0.635	128.1	0.21 ± 0.02	0.18 ± 0.01	1.03	-0.43	0.995
AP	1	1600	0.661	241.6	0.18 ± 0.02	0.17 ± 0.01	0.91	-0.44	0.996
AP	1	3200	0.682	455.9	0.18 ± 0.02	0.15 ± 0.01	0.96	-0.40	0.997
spiral	2	25	0.428	4.8	0.31 ± 0.03	0.31 ± 0.02	1.16	-0.50	1.028
spiral	2	50	0.470	8.7	0.33 ± 0.03	0.30 ± 0.01	1.04	-0.67	1.013
spiral	2	100	0.495	15.6	0.30 ± 0.02	0.29 ± 0.01	0.92	-0.70	1.019
spiral	2	200	0.509	27.9	0.28 ± 0.03	0.30 ± 0.00	0.96	-0.62	1.023
spiral	2	400	0.516	49.9	0.39 ± 0.04	0.32 ± 0.00	1.01	-0.62	0.995
spiral	2	800	0.518	89.7	0.36 ± 0.03	0.33 ± 0.00	1.03	-0.54	1.002
spiral	2	1600	0.519	162.3	0.39 ± 0.04	0.35 ± 0.00	1.11	-0.57	1.005
spiral	2	3200	0.518	295.7	0.33 ± 0.03	0.36 ± 0.00	1.05	-0.53	1.005
random	2	25	0.506	5.9	0.35 ± 0.03	0.13 ± 0.02	1.02	-0.54	0.973
random	2	50	0.520	10.0	0.30 ± 0.03	0.17 ± 0.01	1.09	-0.68	0.991
random	2	100	0.526	17.1	0.33 ± 0.04	0.21 ± 0.01	0.87	-0.72	0.991
random	2	200	0.527	29.5	0.29 ± 0.03	0.25 ± 0.01	0.97	-0.66	0.991
random	2	400	0.526	51.8	0.38 ± 0.04	0.29 ± 0.00	1.06	-0.66	0.992
random	2	800	0.525	91.9	0.33 ± 0.03	0.32 ± 0.00	0.99	-0.56	0.997
random	2	1600	0.523	165.0	0.37 ± 0.03	0.34 ± 0.00	0.99	-0.60	0.998
random	2	3200	0.521	299.1	0.38 ± 0.04	0.36 ± 0.00	0.87	-0.53	0.999
spiral	3	25	0.269	2.9	0.71 ± 0.07	0.65 ± 0.01	1.03	-0.56	0.988
spiral	3	50	0.301	5.3	0.57 ± 0.05	0.62 ± 0.01	1.02	-0.73	0.993
spiral	3	100	0.323	9.6	0.52 ± 0.04	0.60 ± 0.00	0.92	-0.74	1.005
spiral	3	200	0.335	17.4	0.62 ± 0.06	0.59 ± 0.00	0.94	-0.71	1.008
spiral	3	400	0.341	31.4	0.70 ± 0.05	0.59 ± 0.00	0.97	-0.70	1.015
spiral	3	800	0.344	56.7	0.72 ± 0.06	0.59 ± 0.00	1.00	-0.61	1.006
spiral	3	1600	0.345	102.8	0.66 ± 0.07	0.60 ± 0.00	1.01	-0.63	1.011
random	3	25	0.375	4.5	0.55 ± 0.05	0.37 ± 0.01	0.97	-0.58	0.946
random	3	50	0.374	7.2	0.50 ± 0.04	0.42 ± 0.01	1.13	-0.72	0.979
random	3	100	0.372	12.0	0.49 ± 0.04	0.46 ± 0.01	1.10	-0.72	0.971
random	3	200	0.368	20.3	0.50 ± 0.05	0.50 ± 0.00	1.06	-0.70	0.976
random	3	400	0.365	35.1	0.66 ± 0.08	0.53 ± 0.00	0.96	-0.67	0.981
random	3	800	0.362	61.7	0.78 ± 0.12	0.55 ± 0.00	0.90	-0.61	0.987
random	3	1600	0.360	109.9	0.91 ± 0.22	0.57 ± 0.00	0.90	-0.61	0.993
spiral	4	25	0.198	1.9	0.68 ± 0.05	0.76 ± 0.00	0.89	-0.72	0.964
spiral	4	50	0.223	3.6	0.64 ± 0.05	0.74 ± 0.00	1.07	-0.86	0.993
spiral	4	100	0.240	6.5	0.65 ± 0.06	0.72 ± 0.00	0.77	-0.87	0.983
spiral	4	200	0.250	11.9	0.63 ± 0.06	0.71 ± 0.00	1.12	-0.85	0.978
spiral	4	400	0.255	21.5	0.72 ± 0.07	0.71 ± 0.00	1.05	-0.82	1.002
spiral	4	800	0.258	38.9	0.71 ± 0.08	0.71 ± 0.00	1.09	-0.74	1.008
random	4	25	0.299	3.7	0.63 ± 0.06	0.48 ± 0.01	1.06	-0.60	0.973
random	4	50	0.293	5.8	0.63 ± 0.05	0.53 ± 0.01	0.92	-0.71	0.978
random	4	100	0.288	9.5	0.68 ± 0.06	0.58 ± 0.01	1.14	-0.74	0.994
random	4	200	0.284	15.8	0.74 ± 0.06	0.61 ± 0.00	0.89	-0.72	1.006
random	4	400	0.280	27.1	0.75 ± 0.06	0.64 ± 0.00	1.07	-0.69	1.007
random	4	800	0.277	47.3	0.95 ± 0.10	0.66 ± 0.00	1.03	-0.62	1.016
spiral	5	25	0.175	2.0	0.68 ± 0.06	0.79 ± 0.00	0.98	-0.51	1.019
spiral	5	50	0.191	3.6	0.74 ± 0.06	0.77 ± 0.00	1.04	-0.69	0.989
spiral	5	100	0.200	6.4	0.72 ± 0.07	0.76 ± 0.00	1.00	-0.69	0.998
spiral	5	200	0.205	11.3	0.75 ± 0.07	0.76 ± 0.00	1.21	-0.66	1.020
spiral	5	400	0.207	20.3	0.75 ± 0.08	0.76 ± 0.00	1.15	-0.64	1.014
random	5	25	0.247	3.2	0.70 ± 0.06	0.56 ± 0.01	0.93	-0.56	0.954
random	5	50	0.240	5.0	0.76 ± 0.06	0.61 ± 0.01	1.24	-0.70	0.992
random	5	100	0.234	8.0	0.71 ± 0.07	0.65 ± 0.00	0.94	-0.74	1.005
random	5	200	0.230	13.2	0.73 ± 0.07	0.68 ± 0.00	1.02	-0.69	1.021
random	5	400	0.226	22.5	0.70 ± 0.06	0.71 ± 0.00	1.08	-0.68	1.009
spiral	6	25	0.153	1.6	0.74 ± 0.07	0.82 ± 0.00	0.80	-0.57	1.020
spiral	6	50	0.164	2.9	0.79 ± 0.06	0.81 ± 0.00	1.20	-0.73	0.994
spiral	6	100	0.170	5.0	0.71 ± 0.06	0.80 ± 0.00	0.92	-0.76	1.003
spiral	6	200	0.172	8.9	0.75 ± 0.07	0.80 ± 0.00	1.06	-0.70	1.016

family	d	T	ρ	$\bar{\mu}$	Φ	Φ_{HL}	Φ_{rand}	κ	obs/pred
random	6	25	0.212	2.7	0.65 ± 0.05	0.57 ± 0.01	0.93	-0.57	0.904
random	6	50	0.204	4.1	0.59 ± 0.04	0.63 ± 0.01	1.05	-0.74	0.966
random	6	100	0.198	6.4	0.62 ± 0.06	0.68 ± 0.01	0.93	-0.77	0.967
random	6	200	0.194	10.5	0.61 ± 0.05	0.71 ± 0.00	1.09	-0.72	0.990